

A new taxonomy-based protein fold recognition approach based on autocross-covariance transformation

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ABSTRACT

Motivation: Fold recognition is an important step in protein structure and function prediction. Traditional sequence comparison methods fail to identify reliable homologies with low sequence identity, while the taxonomic methods are effective alternatives, but their prediction accuracies are around 70%, which are still relatively low for practical usage.

Results: In this study, a simple and powerful method is presented for taxonomic fold recognition, which combines support vector machine (SVM) with autocross-covariance (ACC) transformation. The evolutionary information represented in the form of position-specific score matrices is converted into a series of fixed-length vectors by ACC transformation and these vectors are then input to a SVM classifier for fold recognition. The sequence-order effect can be effectively captured by this scheme. Experiments are performed on the widely used D-B dataset and the corresponding extended dataset, respectively. The proposed method, called ACCFold, gets an overall accuracy of 70.1% on the D-B dataset, which is higher than major existing taxonomic methods by 2–14%. Furthermore, the method achieves an overall accuracy of 87.6% on the extended dataset, which surpasses major existing taxonomic methods by 9–17%. Additionally, our method obtains an overall accuracy of 80.9% for 86-folds and 77.2% for 199-folds. These results demonstrate that the ACCFold method provides the state-of-the-art performance for taxonomic fold recognition.

Availability: The source code for ACC transformation is freely available at <http://www.iipl.fudan.edu.cn/demo/accpkg.html>.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 INTRODUCTION

The prediction of 3D structure of a protein from its amino acid sequence is one of the most important tasks in computational biology. A solution to this enigma is ever more essential considering that the number of completely sequenced genomes rapidly increases (Wu *et al.*, 2006), while the proteins with known structures are very limited (Kouranov *et al.*, 2006). Theoretical prediction methods provide possible ways to fill in this gap. During the last

decades, many template-based and template-free methods have been presented (Moult *et al.*, 2007). A protein structure can be accurately modeled when the homologous templates are identified. However, the detection of homologies with low sequence identity remains a challenging problem.

Fold recognition has been proven an effective method for protein structure prediction. The general sequence comparison methods such as sequence–sequence alignment (Altschul *et al.*, 1997), sequence–profile alignment (Gough *et al.*, 2001) and the profile–profile alignment (Wang and Dunbrack, 2004) fail to recognize homologous templates when the similarities between the query sequence and the templates are low. An alternative is to thread the query sequence onto the template structures (Wu and Zhang, 2007). Unfortunately, these methods, although widely used, are not able to achieve satisfactory results at the fold level (Cheng and Baldi, 2006). Recently, the taxonomic methods have been developed for protein fold recognition, which classify a sequence into one of the fold classes according to the SCOP architecture (Andreeva *et al.*, 2004). The basic idea is that the fold types in proteins are finite. The total number of folds in globular, water-soluble proteins is estimated at about 1000 (Wolf *et al.*, 2000). Some variant methods have also been proposed for taxonomic fold recognition. Ding and Dubchak (2001) used the support vector machine (SVM) and neural network for multi-class protein fold recognition and achieved an accuracy of 56% on a dataset containing 27 folds. Here, the accuracy refers to the percentage of proteins whose folds have been correctly identified. Other researchers tried to improve prediction performance by either incorporating new features or developing novel algorithms for multi-class classification (Bindewald *et al.*, 2003; Le *et al.*, 2005; Rangwala and Karypis, 2006). Shen and Chou (2006) presented an ensemble classifier called PFP-Pred, which improves the accuracy to 62% on the same dataset used by Ding and Dubchak (2001). Other ensemble classifiers have also been developed, such as the genetic algorithm optimized ensemble classifier (Guo and Gao, 2008), the probabilistic multi-class multi-kernel classifier (Damoulas and Girolami, 2008), the hybrid machine learning classifier (Melvin *et al.*, 2008). Besides the basic sequences or physical–chemical propensities of amino acids as fold discriminatory features, the informations such as secondary structural state and solvent accessibility state can be used to further improve the accuracy to 70% (Chen and Kurgan, 2007; Shamim *et al.*, 2007). The most promising result was achieved by Deschavanne and Tufféry (2009), who employed the hidden Markov

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structure alphabet as additional feature and got an accuracy of 78% for 27 folds (extended D-B dataset) and 68% for 60 folds.

Machine learning-based techniques such as SVM and neural network require fixed-length vectors for training. However, protein sequences often have different lengths. Since protein sequences have univariate direction from beginning to end, which is analogous to time sequences of process data, thus time series models can be used. Here, the autocross covariance (ACC) transformation is introduced to transform protein sequences into fixed-length vectors. As a statistical tool for analyzing sequences of vectors developed by Wold *et al.* (1993), ACC has been successfully used for protein family classification (Guo *et al.*, 2006; Lapinsh *et al.*, 2002) and protein interaction prediction (Guo *et al.*, 2008). Since each residue has many physical-chemical properties, such as hydrophobicity, hydrophilicity, normalized van der Waals volume, polarity, polarizability, sequence profile, etc., a sequence can be represented as a numeric matrix. ACC can measure the correlation of two properties (or the same property) along the protein sequence and transforms the matrix into a fixed-length vector. Note that the PFP-Pred method implicitly used the autocovariance (AC) transformation for hydrophobicity and hydrophilicity properties (Shen and Chou, 2006).

In this study, a novel taxonomic method that combines SVM with ACC is presented for protein fold recognition. Previous successful applications of PSI-BLAST profile show that evolutionary information is more informative than query sequence itself, so the position-specific score matrix (PSSM) is transformed into a fixed-length vector by ACC. This vector is then input to a SVM classifier to predict folds. Experiments are performed on the widely used D-B dataset established by Ding and Dubchak (2001) and the extended D-B dataset, respectively. The proposed method, called ACCFold, gets an overall accuracy of 70.1% on the D-B dataset and 87.6% on the extended D-B dataset, which outperforms major existing taxonomic methods. Furthermore, the proposed method can be applied to large-scale folds without significant performance degradation. Performance comparison with the alignment-based threading methods is also presented.

2 MATERIALS AND METHODS

2.1 Datasets

To evaluate the proposed method and compare it with existing methods, five datasets are used here: the D-B dataset (Ding and Dubchak, 2001), the extended D-B dataset, the F86 and F199 datasets and the Lindahl dataset (Lindahl and Elofsson, 2000).

The D-B dataset (available at <http://ranger.uta.edu/~chqding/protein/>) has been widely used by many existing studies (Chen and Kurgan, 2007; Guo and Gao, 2008; Shamim *et al.*, 2007; Shen and Chou, 2006). It contains 311 proteins for training and 383 proteins for test. According to Ding and Dubchak (2001), any two proteins in the training set share <40% identity and each fold has at least seven members. Each protein in the test set has not more than 35% identity to any protein in the training set. The proteins in this dataset belong to 27 different SCOP folds representing all major structural classes: all α , all β , α/β , $\alpha+\beta$ and small proteins.

The extended D-B dataset is generated by populating additional protein samples. It is based on the Astral SCOP 1.73 release, in which any two sequences have <40% identity. The proteins that belong to any of the 27 folds of the D-B dataset are extracted. The resulting dataset contains 3202 sequences. The fold names and the number of proteins in each fold of the D-B dataset and the extended D-B dataset are listed in Table 1.

Table 1. Summary of the 27 folds in the two datasets

Index	Fold	D-B dataset		Extended D-B dataset
		N_{train}	N_{test}	N
	all α			
1	Globin-like	13	6	41
2	Cytochrome c	7	9	35
3	DNA/RNA-binding	12	20	280
	3-helical bundle			
4	4-helical up-and-down bundle	7	8	66
5	4-helical cytokines	9	9	28
6	EF hand-like	6	9	58
	all β			
7	Immunoglobulin-like	30	44	369
	beta-sandwich			
8	Cupredoxin-like	9	12	47
9	Nucleoplasmin-like/VP (viral coat and capsid proteins)	16	13	60
10	ConA-like	7	6	57
	lectins/glucanases			
11	SH3-like barrel	8	8	124
12	OB-fold	13	19	126
13	Beta-Trefoil	8	4	42
14	Trypsin-like serine proteases	9	4	44
15	Lipocalins	9	7	32
	α/β			
16	TIM beta/alpha-barrel	29	48	322
17	FAD/NAD(P)-binding domain	11	12	67
18	Flavodoxin-like	11	13	124
19	NAD(P)-binding Rossmann	13	27	193
20	P-loop containing NTH	10	12	232
21	Thioredoxin-fold	9	8	108
22	Ribonuclease H-like motif	10	12	120
23	alpha/beta-Hydrolases	11	7	70
24	Periplasmic binding protein-like	11	4	15
	$\alpha+\beta$			
25	beta-grasp (ubiquitin-like)	7	8	117
26	Ferredoxin-like	13	27	302
	Small proteins			
27	Knottins (small inhibitors, toxins, lectins)	13	27	123

The D-B dataset and the extended D-B dataset contain only 27 folds, which cover a small part of the total folds (1086 folds in SCOP 1.73 database). To evaluate their applicabilities, the taxonomic fold recognition methods are often tested on new datasets containing larger numbers of folds. Again, the non-redundant Astral SCOP 1.73 subset is used. Since a small numbers of samples cannot effectively cover the sample space, we select the folds that contains at least N_f (a predefined parameter) members. Two new datasets are derived: F86 and F199. The F86 dataset contains 86 folds and 5671 sequences, in which each fold has at least 25 members. The F199 dataset contains 199 folds and 7437 sequences, in which each fold has at least 10 members. The extended D-B dataset, the F86 and F199 dataset, are available at the Supplementary Material.

The Lindahl dataset (Lindahl and Elofsson, 2000) is used as a benchmark (available at <http://www.bioinfo.se/protein-id/>) to compare the taxonomic fold recognition methods with the threading methods. There are 976 sequences in this dataset and the identity between any two sequences is <40%.

2.2 ACC transformation

Each residue has many physical–chemical properties as mentioned above, so a protein sequence can be viewed as a time sequence of the corresponding properties. In this study, only the evolutionary information represented in the form of PSSM is selected as the feature, since it alone can achieve promising results. The profile of each sequence is generated by running PSI-BLAST (Altschul *et al.*, 1997) against the NCBI's NR dataset (<ftp://ftp.ncbi.nih.gov/blast/db/nr>) with parameters ($-j$ 3, $-e$ 0.01). The element $S_{i,j}$ in the matrix reflects the probability of amino acid i occurring at the position j .

ACC, as one of the multivariate modeling tools, can transform the PSSMs of different lengths into fixed-length vectors by measuring the correlation between any two properties. ACC results in two kinds of variables: AC between the same property, and cross-covariance (CC) between two different properties. The AC variable measures the correlation of the same property between two residues separated by a distance of lg along the sequence, which can be calculated as:

$$AC(i, lg) = \sum_{j=1}^{L-lg} (S_{i,j} - \bar{S}_i)(S_{i,j+lg} - \bar{S}_i) / (L - lg) \quad (1)$$

where i is one of the residues, L is the length of the protein sequence, $S_{i,j}$ is the PSSM score of amino acid i at position j , $\bar{S}_i$ is the average score for amino acid i along the whole sequence:

$$\bar{S}_i = \sum_{j=1}^L S_{i,j} / L \quad (2)$$

In such a way, the number of AC variables can be calculated as $20 * LG$, where LG is the maximum of lg ($lg = 1, 2, \dots, LG$).

The CC variable measures the correlation of two different properties between two residues separated by lg along the sequence, which can be calculated by:

$$CC(i1, i2, lg) = \sum_{j=1}^{L-lg} (S_{i1,j} - \bar{S}_{i1})(S_{i2,j+lg} - \bar{S}_{i2}) / (L - lg) \quad (3)$$

where $i1, i2$ are two different amino acids and $\bar{S}_{i1}$ ($\bar{S}_{i2}$) is the average score for amino acid $i1$ ($i2$) along the sequence. Since the CC variables are not symmetric, the total number of CC variables is $380 * LG$.

In view of ACC transformation, the sequence-order effect can be indirectly and partially, but quite effectively reflected. Additionally, the ACC transformation is independent of sequence alignments.

In this study, each protein sequence is represented as a vector of either AC variable or ACC variable that is a combination of AC and CC.

2.3 Support vector machine

SVM is a class of supervised learning algorithms first introduced by Vapnik (1998). Given a set of labeled training vectors (positive and negative input examples), SVM can learn a linear decision boundary to discriminate the two classes. The result is a linear classification rule that can be used to classify new test examples. When the examples are linearly non-separable, the kernel function can be used to map the examples to a high-order feature space in which the optimal decision boundary can be found. SVM has a strong theoretical foundation of statistical learning and has exhibited excellent performance in practice.

In this study, the LIBSVM package (Chang and Lin, 2009) is used as an implementation of SVM. The radial basis function (RBF) is taken as the

kernel function, which is defined as:

$$K(X_i, X_j) = \exp(-\gamma \|X_i - X_j\|^2) \quad (4)$$

The values of γ and regularization parameter C are optimized on the datasets by cross-validation. For the multi-state classification, the one-versus-one strategy is used.

2.4 Performance metrics

The performance of various methods is evaluated by the overall accuracy, which is the most commonly used metric for assessing the global performance of a multi-class problem. The overall accuracy (Q) is defined as the ratio of correctly predicted samples to all tested samples:

$$Q = \frac{CN}{N} \times 100 \quad (5)$$

where CN is the number of samples whose folds have been correctly predicted and N is the total number of samples in the test dataset.

Other two measures are used to assess the performance of individual class: sensitivity (Sn) and specificity (Sp):

$$Sn = \frac{TP \times 100}{TP + FN} \quad (6)$$

$$Sp = \frac{TP \times 100}{TP + FP} \quad (7)$$

where TP , FN and FP are the number of true positives, false negatives and false positives, respectively.

The sensitivity measures the accuracy for each class, which is the same as the Q_i defined in other studies (Ding and Dubchak, 2001). The overall accuracy can be calculated as the weighted average of the sensitivities of all classes.

3 RESULTS AND DISCUSSIONS

To evaluate the proposed method (ACCFold), we first check the impact of parameter LG on the performance of ACCFold, then conduct an extensive performance comparison between ACCFold and major existing taxonomy-based methods, including LA kernel (Saigo *et al.*, 2004), SWPSSM kernel (Rangwala and Karypis, 2005) and Shamim method (Shamim *et al.*, 2007) over D-B dataset and the extended D-B dataset, F86 and F199 datasets, and finally we compare ACCFold (including some other taxonomy-based methods) with typical threading methods.

For the D-B dataset, a 2-fold cross-validation is performed, where the original training and test sets are used as the sample partitions. For the extended D-B dataset, a 5-fold cross-validation is performed, where the samples in each fold are first randomly divided into five groups, and from each of the 27 folds, one group is randomly collected, and then the five selected groups are merged into a larger group. Thus finally, five large groups are generated for cross-validation. In the cross-validation experiments, the average performance across all trials is reported. The F86 and F199 datasets are used to test the prediction capability of the proposed method in the cases of large numbers of folds.

The LA kernel and the SWPSSM kernel have been successfully used for binary classification of remote homology detection. Here, we extend them to multi-class classification for fold recognition. The Shamim method is one of the best fold recognition methods so far. All these methods are in-house implemented and tested on the same datasets to give an unbiased comparison with the ACCFold method.

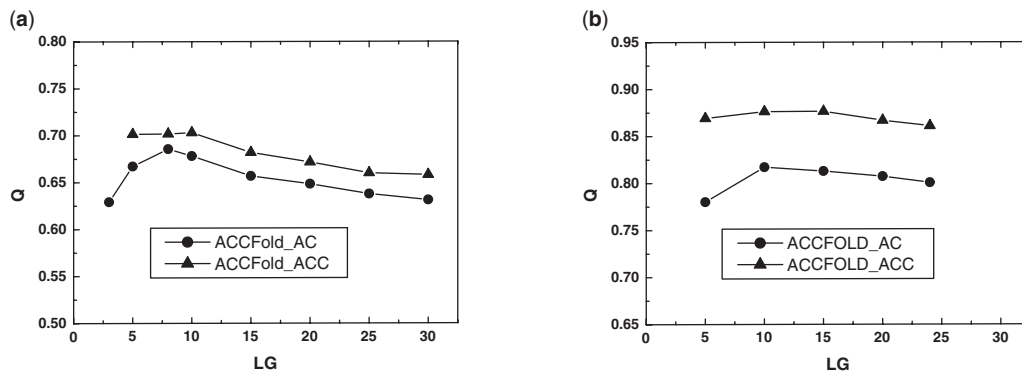


Fig. 1. Overall accuracy versus LG value. Subfigures (a) and (b) are experimental results of the D-B dataset and the extended D-B dataset, respectively.

3.1 The impact of LG

The proposed method can use AC or ACC variables, and the corresponding methods are referred to as ACCFold_AC and ACCFold_ACC, respectively. The value of LG can impact the performance. The maximum value of LG is the length of the shortest sequence minus one, which is 31 for the D-B dataset and 25 for the extended D-B dataset. The overall accuracies for different values of LG are shown in Figure 1. As can be seen, the value of LG has modest influence on the overall accuracy. The accuracy first increases to a maximum value and then slightly goes down as the value of LG increases. On the D-B dataset, the optimal values of LG for AC and ACC are 8 and 10, respectively, whereas, on the extended dataset, the best values of LG for both AC and ACC variables is 10.

3.2 Performance comparison with existing taxonomy-based methods

3.2.1 Results on the D-B dataset The performance comparison with three other taxonomy-based methods by 2-fold cross-validation is shown in Table 2. The detailed results are given in the Supplementary Material. As can be seen, the ACCFold method with AC variable gets an overall accuracy of 68.6%. When the ACC variable is used, the accuracy further increases to 70.1%, which stands for the state-of-the-art performance over this dataset. The folds in the all α and all β classes are easy to be predicted, while the folds in the other three classes are relatively difficult to be discriminated. Most of the folds in the last three classes achieve high specificities but low sensitivities, which indicates that these folds are under predicted, that is, many samples are predicted to be other folds. To the best of our knowledge, except the method (referred to as multi-kernel method here) developed by Damoulas and Girolami (2008), all the other existing methods reported their accuracies not more than 70% over the D-B dataset (Chen and Kurgan, 2007; Ding and Dubchak, 2001; Guo and Gao, 2008; Shen and Chou, 2006). The multi-kernel method is based on a complex Bayesian hierarchical framework that combines multiple kernels and its best result is 70%, obtained by 20 randomly initialized trials. The average accuracy is 68.1% with SD of 1.2%. So the method may be largely dependent on the weight parameters of different kernels. The Shamim method gets an overall accuracy of 59.1%, which is slightly lower than the accuracy reported in the literature (59.5%). Such minor difference may be caused by the different schemes used

Table 2. Performance comparison on the D-B dataset

Methods	Q	ave Sp	ave Sn
ACCFold_ACC	70.1	83.3	69.3
ACCFold_AC	68.6	74.9	68.6
Shamim	59.1	66.9	54.7
LA	38.8	85.3	32.0
SWPSSM	67.8	92.6	63.7

for selecting the SVM parameters. Our current study uses the same optimal parameters to test all the groups during cross-validation, whereas the Shamim method selected optimal parameters for each test group (Supplementary Table II of Shamim *et al.*, 2007). In addition, the Shamim method seems to be not robust, since the best results of different datasets are obtained with different combinations of features (Shamim *et al.*, 2007). The LA method has extremely low performance. This may be due to the low sequence identity in this dataset, noting that the LA kernel measures the sequence similarity using only the sequence information. The SWPSSM kernel achieves attractive performance with an overall accuracy of 67.8%, which uses the PSSM to compute the kernel. Such a result indicates that evolutionary information is more informative than the sequence itself.

To give a more comprehensive comparison, we consider several other methods in the literature. All these methods have been tested on the D-B dataset, and their best accuracies are presented in Table 3. They either use different sequence features or employ different algorithms for multi-class classification. As can be seen from the table, the proposed ACCFold method outperforms these methods by 2–14%.

3.2.2 Results on the extended D-B dataset In the D-B dataset, each fold contains a relatively small number of samples, which is unfavorable to fold prediction. So the extended D-B dataset is used.

The performance comparison with three other taxonomy-based methods by 5-fold cross-validation is shown in Table 4. The detailed results are given in the Supplementary Material. As shown in the table, all the methods get improved performance in comparison with that obtained on the D-B dataset. This may be mainly attributed to the larger number of samples in the extended D-B dataset, from

Table 3. The best overall accuracy comparison with some typical existing taxonomic methods on the D-B dataset

Methods	Q	Source
D-D	56	Ding and Dubchak (2001)
Shamim	60.5	Shamim <i>et al.</i> (2007)
PFP-Pred	62.1	Shen and Chou (2006)
GAOEC	64.5	Guo and Gao (2008)
PFRES	68.4	Chen and Kurgan (2007)
Multi-kernel	68.1 ^a	Damoulas and Girolami (2008)
ACCFold	70.1	This study

^aThe average result after 20 randomly initialized trials.**Table 4.** Performance comparison on the extended D-B dataset

Methods	Q	ave Sp	ave Sn
ACCFold_ACC	87.6	90.8	85.1
ACCFold_AC	81.7	86.3	78.1
Shamim	68.9	73.4	55.0
LA	61.3	69.6	39.9
SWPSSM	86.8	93.3	80.4

which powerful classifiers can be trained. Other related studies have also demonstrated that the usage of extended datasets can improve the performance of protein fold recognition (Deschavanne and Tufféry, 2009; Shamim *et al.*, 2007). The ACCFold method with AC variable achieves an overall accuracy of 81.7%. The usage of ACC variable boosts the accuracy to 87.6%, which is the best result for multi-class protein fold recognition so far. The sequence length has minor influence on the proposed methods (see Supplementary Table 3). Both the short sequences and the long sequences can be quite successfully predicted. All the folds in different classes are better predicted than those of the D-B dataset. Especially, the performance of the folds in the α/β , $\alpha + \beta$ and small proteins classes are significantly improved. This may be attributed to the extra samples in the extended dataset, such that the sample space can be effectively covered. Another interesting observation is that the folds with large numbers of samples must not achieve absolutely high performance. This can be explained as follows. In the training of SVM, only the samples near the decision boundary contribute to the classifiers, while most of the samples, which are far away from the decision boundary, are not used. The Shamim method gets an overall accuracy of 68.9%, which is slightly lower than that reported in the literature (70%) (Shamim *et al.*, 2007). Besides the factor mentioned above, another reason is the difference of datasets. Even the LA method, which uses the sequence alone, has significantly improved performance on the extended dataset with an overall accuracy of 61.3%. The SWPSSM method gets an overall accuracy of 86.8%, which is comparable with the ACCFold method. Both methods use evolution information but in different manners. The SWPSSM method measures the similarity between any two sequences by the Smith–Waterman local alignment algorithm (Smith and Waterman, 1981). The ACCFold method measures the correlation between

Table 5. The best overall accuracy comparison with some typical existing taxonomic methods on the extended D-B dataset

Methods	Q	SCOP version	No. ^a	Source
Shamim	70.5	1.71	2554	Shamim <i>et al.</i> (2007)
Deschavanne	78	1.73	3239	Deschavanne and Tufféry (2009)
ACCFold	87.6	1.73	3202	This study

^aThe total number of proteins in the dataset.**Table 6.** Performance comparison on the F86 and F199 dataset

Methods	F86			F199		
	Q	ave Sp	ave Sn	Q	ave Sp	ave Sn
ACCFold_ACC	80.9	89.4	76.1	77.2	87.7	70.2
ACCFold_AC	75.6	84.2	71.7	70.0	80.1	63.7
SWPSSM	82.5	95.8	77.4	74.3	85.8	64.5
Shamim	39.9	41.8	26.7	32.2	24.1	14.7
LA	44.0	57.1	24.7	33.7	31.7	14.7

any two residues along the sequence, which takes the sequence-order effect into consideration. In addition, the ACCFold method is superior to SWPSSM method in terms of time complexity. Both methods first compute the kernels between any two samples and then input the kernels to SVM for fold recognition. The SWPSSM method is based on the dynamic programming algorithm, which need a time complexity of $O(n^2L^2)$ to compute the kernel, where n is the total number of samples and L is the length of the longest sequence in the dataset. The ACCFold method first converts the sequence into a fixed-length vector and then computes the kernel by the kernel function, which need a time complexity of $O(n \cdot 400 \cdot LG \cdot L) + O(n^2 \cdot 400 \cdot LG) \approx O(nL^2 + n^2L)$.

Some researchers (Deschavanne and Tufféry, 2009; Shamim *et al.*, 2007) also developed extended D-B datasets for protein fold recognition to further improve the performance. Those extended datasets are based on the same 27 folds but use different versions of the SCOP database. The maximum sequence identity between any two sequences is 40%. Their best overall accuracies and ours are listed in Table 5. The proposed ACCFold method gets the highest accuracy of 87.6%, which is higher than the other methods by 9–17%. The outstanding performance of our approach can be attributed to the effective usage of evolutionary information by ACC transformation as well as well-trained fold-specific SVM.

3.2.3 Results on the F86 and F199 datasets The results on these two datasets are listed in Table 6. The ACCFold method with ACC variable gets an overall accuracy of 80.9% on the F86 dataset and 77.2% on the F199 dataset. The results indicate that the proposed method can be applied to the cases of large number of folds without significantly affecting its performance, as long as the number of samples in each fold is not too small. More folds will be covered if the number of sequences in the non-redundant SCOP subset increases, for example, by adopting a larger threshold of the sequence identity. It is possible to employ other

Table 7. Performance comparison with threading methods on the Lindahl dataset

	Family level			Superfamily level			Fold level		
	Subset0 $N_{\min}=1$	Subset1 $N_{\min}=3$	Subset2 $N_{\min}=4$	Subset0 $N_{\min}=1$	Subset1 $N_{\min}=3$	Subset2 $N_{\min}=5$	Subset0 $N_{\min}=1$	Subset1 $N_{\min}=3$	Subset2 $N_{\min}=5$
No. of sequences	555	97	47	434	205	80	321	225	177
No. of categories	176	13	5	86	20	5	38	14	8
ACCFold_ACC	53.9 ^a	79.6	95.7	23.1	55.4	78.3	29.9	41.4	51.9
ACCFold_AC	53.1	79.5	93.6	20.0	47.7	64.0	28.0	41.3	50.9
SWPSSM	40.9	76.2	91.4	15.4	43.0	48.7	28.1	39.8	50.4
Shamim	10.7	40.1	53.4	14.4	43.1	59.8	24.6	36.8	46.2
LA	7.3	34.0	65.8	6.3	20.0	27.7	18.7	26.6	33.9
RAPTOR ^b		86.6			56.3			38.2	
HHPred ^c		82.9			58.8			25.2	
SP3 ^c		81.6			55.3			28.7	
SP5 ^c		82.4			59.8			37.9	
FOLDPro ^c		85.0			55.5			26.5	

^aOnly the top 1 accuracies are reported for all methods.

^bFrom Xu (2005).

^cFrom Zhang *et al.* (2008).

sophisticated techniques to construct more sensitive classifiers for practical applications, such as training samples selection, active learning, incremental learning, etc., which are left to our future work.

3.3 Performance comparison with threading methods

3.3.1 Results on the Lindahl dataset The taxonomic fold recognition methods construct classifiers to discriminate the correct folds of proteins while the threading methods, such as RAPTOR (Xu, 2005; Xu *et al.*, 2003), HHPred (Soding *et al.*, 2005), FOLDPro (Cheng and Baldi, 2006) and SP3/SP5 (Zhang *et al.*, 2008), use the sequence–template alignments to detect the remote homologies of proteins. To compare these two kinds of methods, the Lindahl dataset (Lindahl and Elofsson, 2000) is used as a benchmark. Remote homology is simulated at the family, the superfamily and the fold level, respectively. A 2-fold cross-validation is performed. At the family level, we select the families that contain at least two samples, and the samples in each family are randomly divided into two groups for cross-validation. At the superfamily (or the fold) level, we select the superfamilies (or folds) that contain at least two families (or superfamilies), and the samples of the same family (superfamily) are placed into the same group for cross-validation, so the training samples and the testing samples come from different families (superfamilies). Testing on the whole dataset is unfair for the taxonomic fold recognition methods because many categories at each level contain extremely small numbers of samples. Most of the taxonomic methods cannot construct effective classifiers with so small numbers of samples. So the categories at each level are further selected by specifying that the number of samples in each group should be larger than or equal to a threshold $N_{\min}$.

Table 7 lists the results of five taxonomic fold recognition methods and five threading methods. The results of taxonomic methods are obtained by 2-fold cross-validation, and the results of threading methods are taken from the references. When the minimum number of samples in each category is 2 ($N_{\min}=1$), the performance of the

taxonomic methods is lower than that of the threading methods. As the number of samples increases, the performance of the taxonomic methods get better. Of course, the number of categories decreases as $N_{\min}$ increases, where homology detection seems to be easier for the taxonomic methods.

3.3.2 Results on the F86 dataset Since there are no sufficient samples for most of the categories in the Lindahl dataset, the F86 dataset is used to compare the performance of different fold recognition methods. Experiments are also performed at the family, the superfamily and the fold level. A 2-fold cross-validation is used to get the final results. The scheme of sample segmentation is the same as that used on the Lindahl dataset. At each level, we select the categories that contain at least five samples in each group ($N_{\min}=5$). All the taxonomic fold recognition methods are locally implemented and tested on this dataset. However, most of the threading methods do not have available packages or are difficult to be configured. So we only compare the taxonomic fold recognition methods with the HHpred method (Soding *et al.*, 2005). The results are listed in Table 8. The ACCFold_ACC method outperforms the other taxonomic methods at all levels. In comparison with the HHPred method, the performance of ACCFold_ACC method is slightly lower than that of HHPred method at the family and superfamily level, but higher than that of HHPred method at the fold level. Similar result is gotten on the Lindahl dataset.

Since HHPred is not the top-performance threading method, these results may indicate that the taxonomic methods are not as good as threading methods and they are currently difficult to be applied to practical fold recognition due to the small number of samples in many folds. However, it is generally regarded that the total number of folds are limited. And as the number of proteins with known structure increases, there will be more and more space and chance to exploit the taxonomic methods to develop effective fold cognition system. The proposed ACCFold method, as a new taxonomic protein

Table 8. Performance comparison with threading methods on the F86 dataset

	Family level	Superfamily level	Fold level
No. of sequences	2533	3851	3397
No. of categories	117	92	42
ACCFold_ACC	86.6	43.5	42.2
ACCFold_AC	81.6	34.5	34.4
SWPSSM	86.2	38.9	42.1
Shamim	48.2	20.1	31.3
LA	35.7	14.6	25.1
HHPred	87.3	69.3	37.0

fold recognition method, has shown the state-of-the-art performance, when compared with the existing taxonomy-based methods. Thus, we think it is potentially useful and has considerable academic significance.

4 CONCLUSIONS

In this study, a novel method (ACCFold) that combines SVM with ACC is introduced for taxonomic protein fold recognition. ACC transformation is used to convert the PSSMs into fixed-length vectors, which are input to SVM for fold recognition. Testing on the D-B dataset and the extended D-B dataset has shown that the ACCFold method outperforms existing taxonomic methods. When only 27 folds are considered, the proposed method gets an overall accuracy of 70.1% on the D-B dataset and 87.6% on the extended dataset. Furthermore, the method achieves an overall accuracy of 80.9% for 86 folds and 77.2% for 199 folds. The results obtained here stand for the state-of-the-art performance of taxonomic protein fold recognition. We think that ACC transformation can be a valuable tool for protein-level studies.

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Feature Selection Based on Mutual Information: Criteria of Max-Dependency, Max-Relevance, and Min-Redundancy

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Abstract—Feature selection is an important problem for pattern classification systems. We study how to select good features according to the maximal statistical dependency criterion based on mutual information. Because of the difficulty in directly implementing the maximal dependency condition, we first derive an equivalent form, called minimal-redundancy-maximal-relevance criterion (mRMR), for first-order incremental feature selection. Then, we present a two-stage feature selection algorithm by combining mRMR and other more sophisticated feature selectors (e.g., wrappers). This allows us to select a compact set of superior features at very low cost. We perform extensive experimental comparison of our algorithm and other methods using three different classifiers (naive Bayes, support vector machine, and linear discriminate analysis) and four different data sets (handwritten digits, arrhythmia, NCI cancer cell lines, and lymphoma tissues). The results confirm that mRMR leads to promising improvement on feature selection and classification accuracy.

Index Terms—Feature selection, mutual information, minimal redundancy, maximal relevance, maximal dependency, classification.

1 INTRODUCTION

IN many pattern recognition applications, identifying the most characterizing features (or attributes) of the observed data, i.e., feature selection (or variable selection, among many other names) [30], [14], [17], [18], [15], [12], [11], [19], [31], [32], [5], is critical to minimize the classification error. Given the input data D tabled as N samples and M features $X = \{x_i, i = 1, \dots, M\}$, and the target classification variable c , the feature selection problem is to find from the M -dimensional observation space, R^M , a subspace of m features, R^m , that “optimally” characterizes c .

Given a condition defining the “optimal characterization,” a search algorithm is needed to find the best subspace. Because the total number of subspaces is 2^M , and the number of subspaces with dimensions no larger than m is $\sum_{i=1}^m \binom{M}{i}$, it is hard to search the feature subspace exhaustively. Alternatively, many sequential-search-based approximation schemes have been proposed, including best individual features, sequential forward search, sequential forward floating search, etc., (see [30], [14], [13] for a detailed comparison.).

The optimal characterization condition often means the *minimal classification error*. In an unsupervised situation where the classifiers are not specified, minimal error usually requires the maximal statistical dependency of the target class c on the data distribution in the subspace R^m (and vice versa). This scheme is *maximal dependency* (Max-Dependency).

One of the most popular approaches to realize Max-Dependency is *maximal relevance* (Max-Relevance) feature selection: selecting the features with the highest relevance to the target class c . Relevance is usually characterized in terms of correlation or mutual information, of which the latter is one of the widely used measures to define dependency of variables. In this paper, we focus on the discussion of mutual-information-based feature selection.

Given two random variables x and y , their mutual information is defined in terms of their probabilistic density functions $p(x)$, $p(y)$, and $p(x, y)$:

$$I(x; y) = \iint p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx dy. \quad (1)$$

In Max-Relevance, the selected features x_i are required, individually, to have the largest mutual information $I(x_i; c)$ with the target class c , reflecting the largest dependency on the target class. In terms of sequential search, the m best individual features, i.e., the top m features in the descent ordering of $I(x_i; c)$, are often selected as the m features.

In feature selection, it has been recognized that the combinations of individually good features do not necessarily lead to good classification performance. In other words, “the m best features are not the best m features” [4], [3], [14], [30]. Some researchers have studied indirect or direct means to reduce the redundancy among features¹ (e.g., [4], [14], [19], [15], [22], [12], [5]) and select features with the *minimal redundancy* (Min-Redundancy). For example, in the sequential forward floating search [25], the joint dependency of features on the target class is maximized; as a by-product, the redundancy among features might be reduced. In [12], Jaeger et al. presented a prefiltering method to group variables, thus, redundant variables within each group can be removed. In

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[5], we proposed a heuristic *minimal-redundancy-maximal-relevance* (mRMR) framework to minimize redundancy, and used a series of intuitive measures of relevance and redundancy to select promising features for both continuous and discrete data sets.

Our work in this paper focuses on three issues that have not been touched in earlier work. First, although both Max-Relevance and Min-Redundancy have been intuitively used for feature selection, no theoretical analysis is given on why they can benefit selecting optimal features for classification. Thus, the first goal of this paper is to present a theoretical analysis showing that mRMR is equivalent to Max-Dependency for first-order feature selection, but is more efficient.

Second, we investigate how to combine mRMR with other feature selection methods (such as wrappers [18], [15]) into a two-stage selection algorithm. By doing this, we show that the space of candidate features selected by mRMR is more characterizing. This property of mRMR facilitates the integration of other feature selection schemes to find a compact subset of superior features at very low cost.

Third, through comprehensive experiments we compare mRMR, Max-Relevance, Max-Dependency, and the two-stage feature selection algorithm, using three different classifiers and four data sets. The results show that mRMR and our two-stage algorithm are very effective in a wide range of feature selection applications.

This paper is organized as follows: Section 2 presents the theoretical analysis of the relationships of Max-Dependency, Max-Relevance, and Min-Redundancy. Section 3 presents the two-stage feature selection algorithm, including schemes to integrate wrappers to select a squeezed subset of features. Section 4 discusses implementation issues of density estimation for mutual information and several different classifiers. Section 5 gives experimental results on four data sets, including handwritten characters, arrhythmia, NCI cancer cell lines, and lymphoma tissues. Sections 6 and 7 are discussions and conclusions, respectively.

2 RELATIONSHIPS OF MAX-DEPENDENCY, MAX-RELEVANCE, AND MIN-REDUNDANCY

2.1 Max-Dependency

In terms of mutual information, the purpose of feature selection is to find a feature set S with m features $\{x_i\}$, which jointly have the largest dependency on the target class c . This scheme, called Max-Dependency, has the following form:

$$\max D(S, c), \quad D = I(\{x_i, i = 1, \dots, m\}; c). \quad (2)$$

Obviously, when m equals 1, the solution is the feature that maximizes $I(x_j; c)$ ($1 \leq j \leq M$). When $m > 1$, a simple incremental search scheme is to add one feature at one time: given the set with $m - 1$ features, S_{m-1} , the m th feature can be determined as the one that contributes to the largest increase of $I(S; c)$, which takes the form of (3):

$$\begin{aligned} I(S_m; c) &= \iint p(S_m, c) \log \frac{p(S_m, c)}{p(S_m)p(c)} dS_m dc \\ &= \iint p(S_{m-1}, x_m, c) \log \frac{p(S_{m-1}, x_m, c)}{p(S_{m-1}, x_m)p(c)} dS_{m-1} dx_m dc \\ &= \int \cdots \int p(x_1, \dots, x_m, c) \log \frac{p(x_1, \dots, x_m, c)}{p(x_1, \dots, x_m)p(c)} \\ &\quad dx_1 \cdots dx_m dc. \end{aligned} \quad (3)$$

Despite the theoretical value of Max-Dependency, it is often hard to get an accurate estimation for multivariate density $p(x_1, \dots, x_m)$ and $p(x_1, \dots, x_m, c)$, because of two difficulties in the high-dimensional space: 1) the number of samples is often insufficient and 2) the multivariate density estimation often involves computing the inverse of the high-dimensional covariance matrix, which is usually an ill-posed problem. Another drawback of Max-Dependency is the slow computational speed. These problems are most pronounced for continuous feature variables.

Even for discrete (categorical) features, the practical problems in implementing Max-Dependency cannot be completely avoided. For example, suppose each feature has three categorical states and N samples. K features could have a maximum $\min(3^K, N)$ joint states. When the number of joint states increases very quickly and gets comparable to the number of samples, N , the joint probability of these features, as well as the mutual information, cannot be estimated correctly. Hence, although Max-Dependency feature selection might be useful to select a very small number of features when N is large, it is not appropriate for applications where the aim is to achieve high classification accuracy with a reasonably compact set of features.

2.2 Max-Relevance and Min-Redundancy

As Max-Dependency criterion is hard to implement, an alternative is to select features based on *maximal relevance* criterion (Max-Relevance). Max-Relevance is to search features satisfying (4), which approximates $D(S, c)$ in (2) with the mean value of all mutual information values between individual feature x_i and class c :

$$\max D(S, c), \quad D = \frac{1}{|S|} \sum_{x_i \in S} I(x_i; c). \quad (4)$$

It is likely that features selected according to Max-Relevance could have rich redundancy, i.e., the dependency among these features could be large. When two features highly depend on each other, the respective class-discriminative power would not change much if one of them were removed. Therefore, the following *minimal redundancy* (Min-Redundancy) condition can be added to select mutually exclusive features [5]:

$$\min R(S), \quad R = \frac{1}{|S|^2} \sum_{x_i, x_j \in S} I(x_i, x_j). \quad (5)$$

The criterion combining the above two constraints is called “minimal-redundancy-maximal-relevance” (mRMR) [5]. We define the operator $\Phi(D, R)$ to combine D and R and consider the following simplest form to optimize D and R simultaneously:

$$\max \Phi(D, R), \quad \Phi = D - R. \quad (6)$$

In practice, incremental search methods can be used to find the near-optimal features defined by $\Phi(\cdot)$. Suppose we already have S_{m-1} , the feature set with $m - 1$ features. The task is to select the m th feature from the set $\{X - S_{m-1}\}$. This is done by selecting the feature that maximizes $\Phi(\cdot)$. The respective incremental algorithm optimizes the following condition:

$$\max_{x_j \in X - S_{m-1}} \left[I(x_j; c) - \frac{1}{m-1} \sum_{x_i \in S_{m-1}} I(x_j; x_i) \right]. \quad (7)$$

The computational complexity of this incremental search method is $O(|S| \cdot M)$.

2.3 Optimal First-Order Incremental Selection

We prove in the following that the combination of Max-Relevance and Min-Redundancy criteria, i.e., the mRMR criterion, is equivalent to the Max-Dependency criterion if one feature is selected (added) at one time. We call this type of selection the “first-order” incremental search. We have the following theorem:

Theorem. *For the first-order incremental search, mRMR is equivalent to Max-Dependency (2).*

Proof. By definition of the first-order search, we assume that S_{m-1} , i.e., the set of $m-1$ features, has already been obtained. The task is to select the optimal m th feature x_m from set $\{X - S_{m-1}\}$.

The dependency D in (2) and (3) is represented by mutual information, i.e., $D = I(S_m; c)$, where $S_m = \{S_{m-1}, x_m\}$ can be treated as a multivariate variable. Thus, by the definition of mutual information, we have:

$$\begin{aligned} I(S_m; c) &= H(c) + H(S_m) - H(S_m, c) \\ &= H(c) + H(S_{m-1}, x_m) - H(S_{m-1}, x_m, c), \end{aligned} \quad (8)$$

where $H(\cdot)$ is the entropy of the respective multivariate (or univariate) variables.

Now, we define the following quantity $J(S_m) = J(x_1, \dots, x_m)$ for scalar variables $x_1, \dots, x_m$,

$$\begin{aligned} J(x_1, x_2, \dots, x_m) &= \\ \int \cdots \int p(x_1, \dots, x_m) \log \frac{p(x_1, x_2, \dots, x_m)}{p(x_1) \cdots p(x_m)} dx_1 \cdots dx_m. \end{aligned} \quad (9)$$

Similarly, we define $J(S_m, c) = J(x_1, \dots, x_m, c)$ as

$$\begin{aligned} J(x_1, x_2, \dots, x_m, c) &= \\ \int \cdots \int p(x_1, \dots, x_m, c) \log \frac{p(x_1, x_2, \dots, x_m, c)}{p(x_1) \cdots p(x_m) p(c)} dx_1 \cdots dx_m dc. \end{aligned} \quad (10)$$

We can easily derive (11) and (12) from (9) and (10),

$$H(S_{m-1}, x_m) = H(S_m) = \sum_{i=1}^m H(x_i) - J(S_m), \quad (11)$$

$$H(S_{m-1}, x_m, c) = H(S_m, c) = H(c) + \sum_{i=1}^m H(x_i) - J(S_m, c). \quad (12)$$

By substituting them to the corresponding terms in (8), we have

$$\begin{aligned} I(S_m; c) &= J(S_m, c) - J(S_m) \\ &= J(S_{m-1}, x_m, c) - J(S_{m-1}, x_m). \end{aligned} \quad (13)$$

Obviously, Max-Dependency is equivalent to simultaneously maximizing the first term and minimizing the second term.

We can use the Jensen's Inequality [16] to show the second term $J(S_{m-1}, x_m)$ is lower-bounded by 0. A

related and slightly simpler proof is to consider the inequality $\log(z) \leq z - 1$ with the equality if and only if $z = 1$. We see that

$$\begin{aligned} &-J(x_1, x_2, \dots, x_m) \\ &= \int \cdots \int p(x_1, \dots, x_m) \log \frac{p(x_1) \cdots p(x_m)}{p(x_1, \dots, x_m)} dx_1 \cdots dx_m \\ &\leq \int \cdots \int p(x_1, \dots, x_m) \left[\frac{p(x_1) \cdots p(x_m)}{p(x_1, \dots, x_m)} - 1 \right] dx_1 \cdots dx_m \\ &= \int \cdots \int p(x_1) \cdots p(x_m) dx_1 \cdots dx_m \\ &\quad - \int \cdots \int p(x_1, \dots, x_m) dx_1 \cdots dx_m \\ &= 1 - 1 = 0. \end{aligned} \quad (14)$$

It is easy to verify that the minimum is attained when $p(x_1, \dots, x_m) = p(x_1) \cdots p(x_m)$, i.e., all the variables are independent of each other. As all the $m-1$ features have been selected, this pair-wise independence condition means that the mutual information between x_m and any selected feature x_i ($i = 1, \dots, m-1$) is minimized. This is the Min-Redundancy criterion.

We can also derive the upper bound of the first term in (13), $J(S_{m-1}, c, x_m)$. For simplicity, let us first show the upper bound of the general form $J(y_1, \dots, y_n)$, assuming there are n variables $y_1, \dots, y_n$. This can be seen as follows:

$$\begin{aligned} J(y_1, y_2, \dots, y_n) &= \\ \int \cdots \int p(y_1, \dots, y_n) \log \frac{p(y_1, \dots, y_n)}{p(y_1) \cdots p(y_n)} dy_1 \cdots dy_n \\ &= \int \cdots \int p(y_1, \dots, y_n) \log \frac{p(y_1|y_2, \dots, y_n) p(y_2|y_3, \dots, y_n) \cdots p(y_{n-1}|y_n) p(y_n)}{p(y_1) \cdots p(y_{n-1}) p(y_n)} dy_1 \cdots dy_n \\ &= \sum_{i=1}^{n-1} H(y_i) - H(y_1|y_2, \dots, y_n) - H(y_2|y_3, \dots, y_n) - \cdots - H(y_{n-1}|y_n) \\ &\leq \sum_{i=1}^{n-1} H(y_i). \end{aligned} \quad (15)$$

Equation (15) can be easily extended as

$$\begin{aligned} J(y_1, y_2, \dots, y_n) &\leq \min \\ \left\{ \sum_{i=2}^n H(y_i), \sum_{i=1, i \neq 2}^n H(y_i), \dots, \sum_{i=1, i \neq n-1}^n H(y_i), \sum_{i=1}^{n-1} H(y_i) \right\}. \end{aligned} \quad (16)$$

It is easy to verify the maximum of $J(y_1, \dots, y_n)$ or, similarly, the first term in (13), $J(S_{m-1}, c, x_m)$, is attained when all variables are maximally dependent. When S_{m-1} has been fixed, this indicates that x_m and c should have the maximal dependency. This is the Max-Relevance criterion.

Therefore, according to (13), as a combination of Max-Relevance and Min-Redundancy, mRMR is equivalent to Max-Dependency for first-order selection. $\square$

Note that the quantity $J(\cdot)$ in (9) and (10) has also been called “mutual information” for multiple scalar variables [10]. We have the following observations:

1. Minimizing $J(S_m)$ only is equivalent to searching mutually exclusive (independent) features.² This is insufficient for selecting highly discriminative features.
2. Maximizing $J(S_m, c)$ only leads to Max-Relevance. Clearly, the difference between mRMR and Max-Relevance is rooted in the different definitions of dependency (in terms of mutual information). Equation (10) does not consider the joint effect of features on the target class. On the contrary, Max-Dependency ((2) and (3)) considers the dependency between the data distribution in subspace R^m and the target class c . This difference is critical in many circumstances.
3. The equivalence between Max-Dependency and mRMR indicates mRMR is an *optimal* first-order implementation scheme of Max-Dependency.
4. Compared to Max-Dependency, mRMR avoids the estimation of multivariate densities $p(x_1, \dots, x_m)$ and $p(x_1, \dots, x_m, c)$. Instead, calculating the bivariate density $p(x_i, x_j)$ and $p(x_i, c)$ could be much easier and more accurate. This also leads to a more efficient feature selection algorithm.

3 FEATURE SELECTION ALGORITHMS

Our goal is to design efficient algorithms to select a compact set of features. In Section 2, we propose a fast mRMR feature selection scheme (7). A remaining issue is how to determine the optimal number of features m . Since a mechanism to remove potentially redundant features from the already selected features has not been considered in the incremental selection, according to the idea of mRMR, we need to refine the results of incremental selection.

We present a two-stage feature selection algorithm. In the first stage, we find a candidate feature set using the mRMR incremental selection method. In the second stage, we use other more sophisticated schemes to search a compact feature subset from the candidate feature set.

3.1 Selecting the Candidate Feature Set

To select the candidate feature set, we compute the cross-validation classification error for a large number of features and find a relatively stable range of small error. This range is called Ω . The optimal number of features (denoted as n^*) of the candidate set is determined within Ω . The whole process includes three steps:

1. Use mRMR incremental selection (7) to select n (a preset large number) sequential features from the input X . This leads to n sequential feature sets $S_1 \subset S_2 \subset \dots \subset S_{n-1} \subset S_n$.
2. Compare all the n sequential feature sets $S_1, \dots, S_k, \dots, S_n$, ($1 \leq k \leq n$) to find the range of k , called Ω , within which the respective (cross-validation classification) error e_k is consistently small (i.e., has both small mean and small variance).
3. Within Ω , find the smallest classification error $e^* = \min e_k$. The optimal size of the candidate feature set, n^* , is chosen as the smallest k that corresponds to e^* .

2. In the field of feature extraction, minimizing $J(S_m)$ has led to an algorithm of independent component analysis [10].

3.2 Selecting Compact Feature Subsets

Many sophisticated schemes can be used to search the compact feature subsets from the candidate set S_{n^*} . To illustrate that mRMR can produce better candidate features, which favors better combination with other methods, we use wrappers to search the compact feature subsets.

A wrapper [15], [18] is a feature selector that convolves with a classifier (e.g., naive Bayes classifier), with the direct goal to minimize the classification error of the particular classifier. Usually, wrappers can yield high classification accuracy for a particular classifier at the cost of high computational complexity and less generalization of the selected features on other classifiers. This is different from the mRMR method introduced above, which does not optimize the classification error directly. The latter type of approach (e.g., mRMR and Max-Relevance), sometimes called “filter” [18], [15], often selects features by testing whether some preset conditions about the features and the target class are satisfied. In practice, the filter approach has much lower complexity than wrappers; the features thus selected often yield comparable classification errors for different classifiers, because such features often form intrinsic clusters in the respective subspace.

By using mRMR feature selection in the first-stage, we intend to find a small set of candidate features, in which the wrappers can be applied at a much lower cost in the second-stage. We will continue our discussion on this point in Section 3.3.

In this paper, we consider two selection schemes of wrapper, i.e., the backward and forward selections:

1. The backward selection tries to exclude one redundant feature at a time from the current feature set S_k (initially, k is set to n^* obtained in Section 3.1), with the constraint that the resultant feature set S_{k-1} leads to a classification error e_{k-1} no worse than e_k . Because every feature in S_k can be considered in removal, there are k different configurations of S_{k-1} . For each possible configuration, the respective classification error e_{k-1} is calculated. If, for every configuration, the corresponding e_{k-1} is larger than e_k , there is no gain in either classification accuracy or feature dimension reduction (i.e., every existing feature in S_k appears to be useful), thus, the backward selection terminates (accordingly, the size of the compact feature subset, m , is set to k). Otherwise, among the k configurations of S_{k-1} , the one that leads to the largest error reduction is chosen as the new feature set. If there are multiple configurations leading to the same error reduction, one of them is chosen randomly. This decremental selection procedure is repeated until the termination condition is satisfied.
2. The forward selection tries to select a subset of m features from S_{n^*} in an incremental manner. Initially, the classification error is set to the number of samples, i.e., N . The wrapper first searches for the feature subset with one feature, denoted as Z_1 , by selecting the feature x_1^* that leads to the largest error reduction. Then, from the set $\{S_n - Z_1\}$, the wrapper selects the feature x_2^* so that the feature set $Z_2 = \{Z_1, x_2^*\}$ leads to the largest error reduction. This incremental selection repeats until the classification error begins to increase, i.e., $e_{k+1} > e_k$. Note that we

allow the incremental search to continue when e_{k+1} equals e_k , because we want to search a space as large as possible. Once the termination condition is satisfied, the selected number of features, m , is chosen as the dimension for which the lowest error is first reached. For example, suppose the sequence of classification errors of the first six features is [10, 8, 4, 4, 4, 7]. The forward selection will terminate at five features, but only return the first three features as the result; in this way we obtain a more compact set of features that minimizes the error.

3.3 Characteristic Feature Space

Given two feature sets S_n^1 and S_n^2 both containing n features, and a classifier Γ , we say the feature space of S_n^1 is more *characteristic* if the classification error (using classifier Γ) on S_n^1 is smaller than on S_n^2 . This definition of characteristic space can be extended recursively to the subsets (subspaces) of S_n^1 and S_n^2 . Suppose we have a feature selection method F to generate a series of feature subsets in $S_n^1 : S_1^1 \subset S_2^1 \subset \dots \subset S_k^1 \subset \dots \subset S_{n-1}^1 \subset S_n^1$, and, similarly, a series of subsets in $S_n^2 : S_1^2 \subset \dots \subset S_k^2 \subset \dots \subset S_n^2$. We say S_n^1 is *recursively more characteristic* (RM-characteristic) than S_n^2 on the range $\Omega = [k_{lower}, k_{upper}] (1 \leq k_{lower} < k_{upper} \leq n)$, if for every $k \in \Omega$, the classification error on S_k^1 is consistently smaller than on S_k^2 .

To determine which one of the feature sets S_n^1 and S_n^2 is superior, it is often insufficient to compare the classification errors for a specific size of the feature sets. A better way is to observe which set is RM-characteristic for a reasonably large range Ω . In the extreme case, we use $\Omega = [1, n]$. Given two feature selection methods F^1 and F^2 , if, the feature sets generated by F^1 are RM-characteristic than those generated by F^2 , we believe the method F^1 is better than F^2 .

Let us consider the following example to compare mRMR and Max-Relevance based on the concept of RM-characteristic feature space. As a comprehensive study, we consider both the sequential and nonsequential feature sets as follows (more details will be given in experiments):

1. A direct comparison is to examine whether the mRMR sequential feature sets are RM-characteristic than Max-Relevance sequential feature sets. We use both methods to select n sequential feature sets $S_1 \subset \dots \subset S_k \subset \dots \subset S_n$ and compute the respective classification errors. If, for most $k \in [1, n]$, we obtain smaller errors on mRMR feature sets, we can conclude that mRMR is better than Max-Relevance for the sequential (or incremental) feature selection.
2. We also use other feature selection methods (e.g., wrappers) in the second stage of our feature-selection algorithm to probe whether mRMR is better than Max-Relevance for nonsequential feature sets. For example, for the mRMR and Max-Relevance candidate feature sets with n^* features, we use the backward-selection-wrapper to produce two series of feature sets with $k = n^* - 1, n^* - 2, \dots, m$ features by removing some nonsequential features that are potentially redundant. Then, the respective classification errors of these feature sets are computed. If, for most k , we find the mRMR nonsequential feature subset leads to lower error, we conclude the mRMR candidate feature set is (approximately) RM-characteristic than the Max-Relevance candidate feature set.

3. Both the forward and backward selections of wrapper are used. Different classifiers (as discussed later in Section 4.2) are also considered in wrappers. We use both mRMR and Max-Relevance methods to select the same number of candidate features and compare the classification errors of the feature subsets thereafter selected by wrappers. If all the observations agree that the mRMR candidate feature set is RM-characteristic, we have high confidence that mRMR is a superior feature selection method.
4. Given two feature sets, if S_n^1 is RM-characteristic than S_n^2 , then it is faithful to compare the lowest errors obtained for the subsets of S_n^1 and S_n^2 .

Clearly, for feature spaces containing the same number of features, wrappers can be applied more effectively on the space that is RM-characteristic. This also indicates that wrappers can be applied at a lower cost, by improving the characterizing strength of features and reducing the number of pre-selected features.

In real situations, it might not be possible to obtain $e_k^1 < e_k^2$ for every k in Ω . Hence, we can define a confidence score $0 \leq \rho \leq 1$ to indicate the percentage of different k values for which the $e_k^1 < e_k^2$ condition is satisfied. For example, when $\rho = 0.90$ (90 percent k -values correspond to the $e_k^1 < e_k^2$ condition), it is safe to claim that S_n^1 is *approximately* RM-characteristic than S_n^2 on Ω . As can be seen in the experiments, usually this approximation is sufficient to compare two series of feature subsets.

4 IMPLEMENTATION ISSUES

Before presenting the experimental results in Section 5, we discuss two implementation issues regarding the experiments: 1) calculation of mutual information for both discrete and continuous data and 2) multiple types of classifiers used in our experiments.

4.1 Mutual Information Estimation

We consider mutual-information-based feature selection for both discrete and continuous data. For discrete (categorical) feature variables, the integral operation in (1) reduces to summation. In this case, computing mutual information is straightforward, because both joint and marginal probability tables can be estimated by tallying the samples of categorical variables in the data.

However, when at least one of variables x and y is continuous, their mutual information $I(x; y)$ is hard to compute, because it is often difficult to compute the integral in the continuous space based on a limited number of samples. One solution is to incorporate data discretization as a preprocessing step. For some applications where it is unclear how to properly discretize the continuous data, an alternative solution is to use density estimation method (e.g., Parzen windows) to approximate $I(x; y)$, as suggested by earlier work in medical image registration [7] and feature selection [17].

Given N samples of a variable x , the approximate density function $\hat{p}(x)$ has the following form:

$$\hat{p}(x) = \frac{1}{N} \sum_{i=1}^N \delta(x - x^{(i)}, h), \quad (17)$$

where $\delta(\cdot)$ is the Parzen window function as explained below, $x^{(i)}$ is the i th sample, and h is the window width. Parzen has

proven that, with the properly chosen $\delta(\cdot)$ and h , the estimation $\hat{p}(x)$ can converge to the true density $p(x)$ when N goes to infinity [21]. Usually, $\delta(\cdot)$ is chosen as the Gaussian window:

$$\delta(z, h) = \exp\left(-\frac{z^T \Sigma^{-1} z}{2h^2}\right) / \left\{ (2\pi)^{d/2} h^d |\Sigma|^{1/2} \right\}, \quad (18)$$

where $z = x - x^{(i)}$, d is the dimension of the sample x and Σ is the covariance of z . When $d = 1$, (17) returns the estimated marginal density; when $d = 2$, we can use (17) to estimate the density of bivariate variable (x, y) , $p(x, y)$, which is actually the joint density of x and y . For the sake of robust estimation, for $d \geq 2$, Σ is often approximated by its diagonal components.

4.2 Multiple Classifiers

Our mRMR feature selection method does not convolve with specific classifiers. Therefore, we expect the features selected by this scheme have good performance on various types of classifiers. To test this, we consider three widely used classifiers, i.e., Naive Bayes (NB), Support Vector Machine (SVM), and Linear Discriminant Analysis (LDA).

NB [20] is one of the oldest classifiers. It is based on the Bayes rule and assumes that feature variables are independent of each other given the target class. Given a sample $s = \{x_1, x_2, \dots, x_m\}$ for m features, the posterior probability that s belongs to class c_k is

$$p(c_k | s) \propto \prod_{i=1}^m p(x_i | c_k), \quad (19)$$

where $p(x_i | c_k)$ is the conditional probability table (or densities) learned from examples in the training process. The Parzen-window density-approximation in (17) and (18) can be used to estimate $p(x_i | c_k)$ for continuous features. Despite the conditional independence assumption, NB has been shown to have good classification performance for many real data sets, on par with many more sophisticated classifiers [20].

SVM [29], [2] is a more modern classifier that uses kernels to construct linear classification boundaries in higher dimensional spaces. We use the LIBSVM package [9], which supports both 2-class and multiclass classification.

As one of the earliest classifiers, LDA [30] learns a linear classification boundary in the input feature space. It can be used for both 2-class and multiclass problems.

5 EXPERIMENTS

We tested our feature selection approach on two discrete and two continuous data sets. For these data sets, we used multiple ways to calculate the mutual information and tested the performance of the selected features based on three classifiers introduced above. In this way, we provided a comprehensive study on the performance of our feature selection approach under different conditions.

This section is organized as follows: After a brief introduction of data sets in Section 5.1, we compare mRMR against Max-Dependency in terms of both feature selection complexity and feature classification accuracy in Section 5.2. These results demonstrate the practical advantages of our mRMR scheme and provide a direct verification of the theoretical analysis in Section 2. Then, in Sections 5.3 and 5.4, we show a detailed comparison of mRMR and Max-Relevance, the latter

of which has been widely used in practice. We do not show the comparison of mRMR with Min-Redundancy since Min-Redundancy alone usually leads to poor classification (and is seldom used to select features in real applications). Due to the space limitation, in the following, we always demonstrate our comprehensive study with the most representative results. For simplicity, we use MaxDep to denote Max-Dependency and MaxRel to denote Max-Relevance throughout the figures, tables, and texts in this section.

5.1 Data Sets

The four data sets we used are shown in Table 1. They have been extensively used in earlier studies [1], [13], [26], [27], [5]. The first two data sets, HDR-MultiFeature (HDR) and Arrhythmia (ARR), are also available on the UCI machine learning archive [28]. The latter two, NCI and Lymphoma (LYM), are available on the respective authors' Web sites. All the raw data are continuous. Each feature variable in the raw data was preprocessed to have zero mean-value and unit variance (i.e., transformed to their z-scores). To test our approaches on both discrete and continuous data, we discretized the first two data sets, HDR and ARR. The other two data sets, NCI and LYM, were directly used for continuous feature selection.

The data set HDR [6], [14], [28] contains 649 features for 2,000 handwritten digits. The target class has 10 states, each of which has 200 samples. To discretize the data set, each feature variable was binarized at the mean value, i.e., it takes 1 if it is larger than the mean value and -1 otherwise. We selected and evaluated features using 10-fold Cross-Validation (CV).

The data set ARR [28] contains 420 samples of 278 features. The target class has two states with 237 and 183 samples, respectively. Each feature variable was discretized into three states at the positions $\mu \pm \sigma$ (μ is the mean value and σ the standard deviation): it takes -1 if it is less than $\mu - \sigma$, 1 if larger than $\mu + \sigma$, and 0 if otherwise. We used 10-fold CV for feature selection and testing.

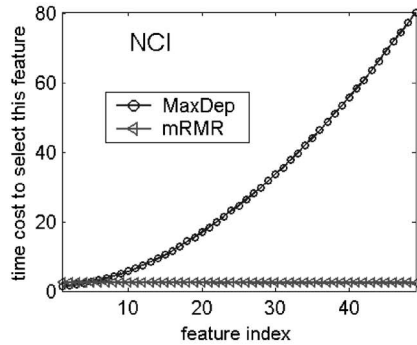
The data set NCI [26], [27] contains 60 samples of 9,703 genes; each gene is regarded as a feature. The target class has nine states corresponding to different types of cancer; each type has two to nine samples. Since the sample number is small, we used the Leave-One-Out (LOO) CV method in testing.

The data set LYM [1] has 96 samples of 4,026 gene features. The target class corresponds to nine subtypes of the lymphoma. Each subtype has two to 46 samples. The sample numbers for these subtypes are highly skewed, which makes it a hard classification problem.

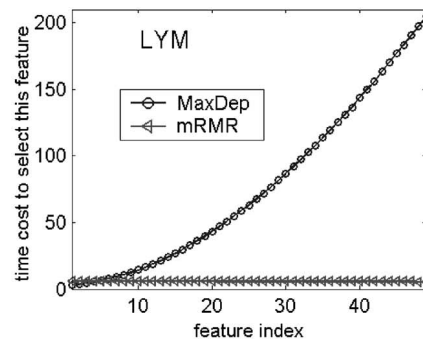
Note that the feature numbers of these data sets are large (e.g., NCI has nearly 10,000 features). These data sets represent some real applications where expensive feature selection methods (e.g., exhaustive search) cannot be used directly. They differ greatly in sample size, feature number, data type (discrete or continuous), data distribution, and target class type (multiclass or 2-class). In addition, we studied different mutual information calculation schemes for both discrete and continuous data and provided results using different classifiers and different wrapper selection schemes. We believe these data and methods provide a comprehensive testing suit for feature selection methods under different conditions.

TABLE 1
Data Sets Used in Our Experiments

Data set	HDR MultiFeat		Arrhythmia		NCI		Lymphoma	
Acronym	HDR		ARR		NCI		LYM	
Source	UCI [28], Duin et al [6][14][13]		UCI [28]		Ross et al [26] Scherf et al [27]		Alizadeh et al [1]	
Raw data type	Continuous							
Experimental data type	Discrete				Continuous			
Processing method	Binarize at μ		Discretize at $\mu \pm \sigma$ to be 3-state		z-score (mean value 0, standard deviation 1)			
# Variable	649		278		9703		4026	
# Sample	2000		420		60		96	
# Class	10		2		9		9	
Class	Name	# Sample	Name	# Sample	Name	# Sample	Name	# Sample
C1	0	200	Normal	237	NSCLC	9	DLBCL	46
C2	1	200	Abnormal	183	Renal	9	CLL	11
C3	2	200			Breast	8	ABB	10
C4	3	200			Melanoma	8	FL	9
C5	4	200			Colon	7	RAT	6
C6	5	200			Leukemia	6	TCL	6
C7	6	200			Ovarian	6	RBB	4
C8	7	200			CNS	5	GCB	2
C9	8	200			Prostate	2	LNT	2
C10	9	200						
Testing method	10-fold CV				LOOCV			



(a)



(b)

Fig. 1. Time cost (seconds) for selecting individual features based on mutual information estimation for continuous data sets. (a) Time cost for selecting each NCI feature. (b) Time cost for selecting each LYM feature.

5.2 Comparison of mRMR and Max-Dependency

The mRMR scheme is a first-order approximation of the Max-Dependency (or MaxDep) selection method. We compared their performances in terms of both feature selection complexity and feature classification accuracy. These comparisons indicate their applicability for real data.

5.2.1 Feature Selection Complexity

In practice, for categorical feature variables, we can introduce an intermediate “joint-feature” variable for MaxDep selection, so that the complexity would not increase much in selecting additional features (the comparison results against mRMR are omitted due to space limitation). Unfortunately, for continuous feature variables, it is hard to adopt a similar approach. For example, we

compared the average computational time cost to select the top 50 mRMR and MaxDep features for both continuous data sets NCI and LYM, based on parallel experiments on a cluster of eight 3.06G Xeon CPUs running Redhat Linux 9, with the Matlab implementation.

The results in Fig. 1 demonstrate that the time cost for MaxDep to select a single feature is a polynomial function of the number of features, whereas, for mRMR, it is almost constant. For example, for NCI data, MaxDep takes about 20 and 60 seconds to select the 20th and 40th features, respectively. In contrast, mRMR always takes about 2 seconds to select any features. For the LYM data, MaxDep needs more than 200 seconds to find the 50th feature, while mRMR uses only 5 seconds. We can conclude that mRMR is computationally much more efficient than MaxDep.

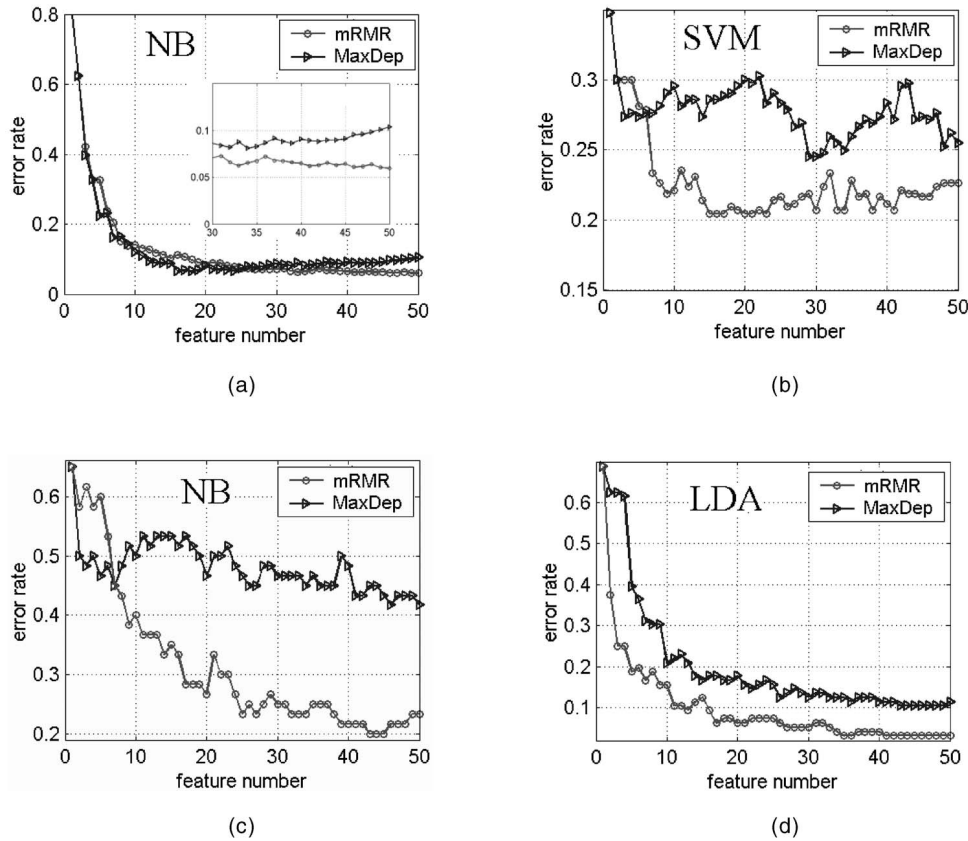


Fig. 2. Comparison of feature classification accuracies of mRMR and MaxDep. Classifiers NB, SVM, and LDA were used. (a) 10-fold CV error rates of HDR data. (b) 10-fold CV error rates of ARR data. (c) LOOCV error rates of NCI data. (d) LOOCV error rates of LYM data.

5.2.2 Feature Classification Accuracy

The selected features for the four data sets were tested using all the three classifiers introduced in Section 4.2. However, for both clarity and brevity, we only plot several representatives of the cross-validation classification error-rate curves in Fig. 2. Similar results were obtained in other cases.

Fig. 2a shows that, for the HDR data, the overall performance of MaxDep and mRMR is similar. MaxDep gets slightly lower errors when the feature number is relatively small, within the range between 1 and 20. When the feature number is larger than 30, the MaxDep features lead to a significantly greater error rate than mRMR features, as indicated in the blow-up windows. For example, 50 mRMR features lead to 6 percent error, in contrast to the 11 percent error of 50 MaxDep features. Noticeably, the two error-rate curves have distinct tendency. For mRMR, the error rate constantly decreases and then converges at some point. On the contrary, the error rate for MaxDep declines for small feature-numbers and then starts to increase for greater feature-numbers, indicating that more features lead to worse classification.

Figs. 2b, 2c, and 2d show the respective comparison results for ARR, NCI, and LYM data sets. The different tendency of the mRMR and MaxDep error-rate curves can be seen more prominently for these three data sets. For example, in Fig. 2b, we see that only with the first three to five features, MaxDep has a slightly lower error rate than mRMR, but the respective error rates are far away from optimum. For all the other feature numbers, mRMR features lead to consistently lower errors than MaxDep. For NCI data in Fig. 2c, MaxDep is better than mRMR only when less than seven features are

used, but its overall classification accuracies are very poor. For a larger feature number, mRMR features lead to only half of the error rate of MaxDep features, indicating a greater discriminating strength. For LYM data in Fig. 2d, MaxDep features are never better than mRMR features.

Why does mRMR tend to outperform MaxDep when the feature number is relatively large? This is because, in high-dimensional space, the estimation of mutual information becomes much less reliable than in two-dimensional space, especially when the number of data samples is comparatively close to the number of joint states of features. This phenomenon is seen more clearly for continuous feature variables, i.e., the NCI and LYM data sets.

This also explains why, for HDR data, the difference between mRMR and MaxDep is not as prominent as those of the three other data sets. Because the HDR data set has a much larger number of data samples than ARR, NCI, and LYM data sets, the accuracy of mutual information estimation for HDR data does not degrade as quickly as those for the other three data sets.

Since the complexity of MaxDep in selecting features is higher and the classification accuracy using MaxDep features is lower, it is much more appealing to make use of mRMR instead of MaxDep in practical feature selection applications. In the following sections, we will focus on comparing mRMR against the most widely used MaxRel selection method.

5.3 Comparison of Candidate Features Selected by mRMR and MaxRel

MaxRel and mRMR have similar computational complexity. The mRMR method is a little bit more expensive, but the

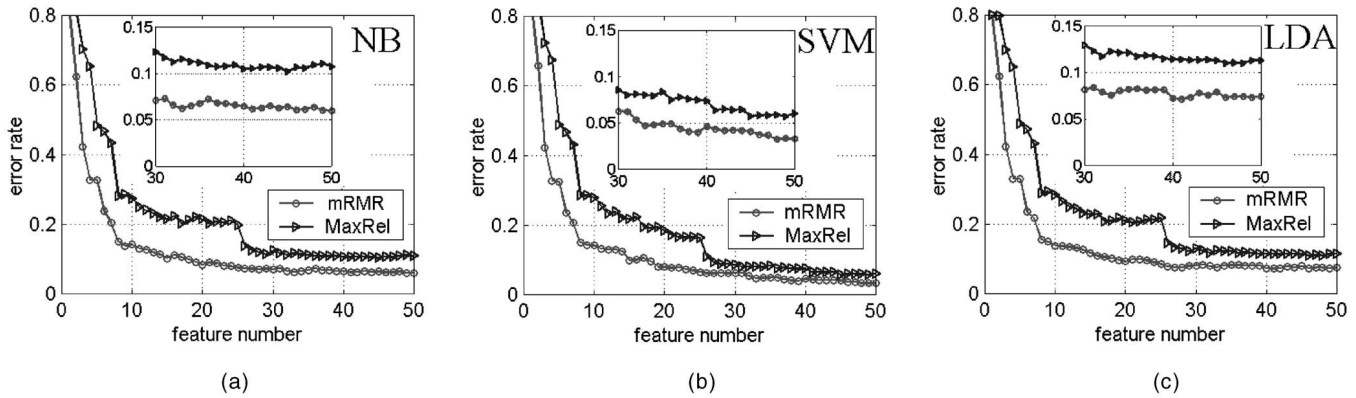


Fig. 3. 10-fold CV error rates of HDR data using mRMR and MaxRel features. (a) NB classifier error rate. (b) SVM classifier error rate. (c) LDA classifier error rate.

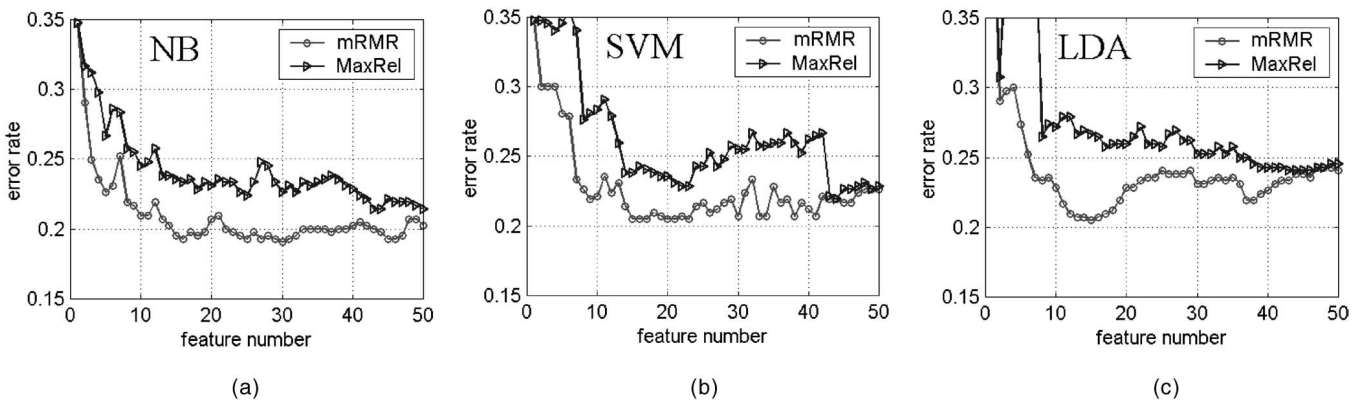


Fig. 4. 10-fold CV error rates of ARR data using mRMR and MaxRel features. (a) NB classifier error rate. (b) SVM classifier error rate. (c) LDA classifier error rate.

difference is minor. Thus, we focus on comparing the feature classification accuracies.

5.3.1 Discrete Data

Figs. 3 and 4 show results of the incremental feature selection and classification for discrete data sets. The feature number ranges from 1 to 50.

For HDR data set, Figs. 3a, 3b, and 3c show the classification error rates with classifiers NB, SVM, and LDA, respectively. Clearly, features selected by mRMR consistently attain significantly lower error rates than those selected by MaxRel. In other words, feature sets selected by mRMR are RM-characteristic than those selected by MaxRel. In this case, it is faithful to compare the lowest classification errors obtained for both methods. As illustrated in the zoom-in windows, with NB, the lowest error rate of mRMR is about 6 percent, while that of MaxRel is about 10 percent; with SVM, the lowest error rate of mRMR is about 3.5 percent, the lowest error rate of MaxRel is about 5.5 percent; with LDA, the lowest error rate of mRMR features is around 7 percent, whereas that of MaxRel is around 11 percent.

Figs. 4a, 4b, and 4c show the classification error rates for the ARR data. Similar to those of the HDR data, features selected by mRMR significantly and consistently outperform those selected by MaxRel. When nearly 50 features are used, the performance of mRMR and MaxRel become close. Overall, the performance of mRMR is much better than that of MaxRel, since 15 mRMR features lead to better classification accuracy than 50 MaxRel features.

Results in this section show that, for discrete data sets, the candidate features selected by mRMR are significantly better than those selected by MaxRel. These effects are independent of the concrete classifiers we used.

5.3.2 Continuous Data

Tables 2 and 3 show the results of the incremental feature selection and classification for continuous data sets. The feature number ranges from 1 to 50 (to save space, we only list results of 1, 5, 10, 15, ..., 50 features).

Table 2 shows that, for NCI data, features selected by mRMR lead to lower error rates than those selected by MaxRel. The differences are consistent and significant. For example, with NB and more than 40 features (for simplicity, this combination is called "NB+40features"), we obtained an error rate around 20 percent for mRMR and around 33 percent for MaxRel. With SVM+40features, we obtained the error rate 23-26 percent for mRMR, and 35-38 percent for MaxRel. With LDA+40features, the results are similar.

Table 3 shows that, for LYM data, mRMR features are also superior to MaxRel features (e.g., 3 percent versus 15 percent for LDA+50features).

Results in this section show that, for continuous data, mRMR also outperforms MaxRel in selecting RM-characteristic sequential feature sets. They also indicate that the Parzen-window-based density-estimation for mutual information computation can be effectively used for feature selection.

TABLE 2
LOOCV Error Rate (%) of NCI Data Using mRMR and MaxRel Features

Classifier	m Method	1	5	10	15	20	25	30	35	40	45	50
NB	MaxRel	65.00	51.67	51.67	45.00	46.67	43.33	41.67	38.33	36.67	33.33	36.67
	mRMR	65.00	60.00	40.00	35.00	26.67	23.33	25.00	25.00	21.67	20.00	23.33
SVM	MaxRel	98.33	46.67	55.00	50.00	45.00	55.00	41.67	35.00	38.33	35.00	36.67
	mRMR	98.33	70.00	58.33	48.33	40.00	31.67	31.67	31.67	26.67	23.33	23.33
LDA	MaxRel	73.33	60.00	60.00	50.00	46.67	46.67	41.67	36.67	38.33	41.67	40.00
	mRMR	73.33	66.67	50.00	53.33	45.00	33.33	35.00	35.00	33.33	30.00	30.00

TABLE 3
LOOCV Error Rate (%) of LYM Data Using mRMR and MaxRel Features

Classifier	m Method	1	5	10	15	20	25	30	35	40	45	50
NB	MaxRel	72.92	25.00	15.63	13.54	13.54	12.50	13.54	12.50	11.46	11.46	10.42
	mRMR	72.92	17.71	16.67	10.42	11.46	9.38	10.42	9.38	9.38	7.29	8.33
SVM	MaxRel	42.71	27.08	21.88	21.88	18.75	16.67	14.58	14.58	15.63	11.46	12.50
	mRMR	42.71	11.46	10.42	7.29	5.21	7.29	7.29	5.21	5.21	5.21	4.17
LDA	MaxRel	68.75	32.29	22.92	23.96	23.96	21.88	22.92	17.71	16.67	16.67	15.63
	mRMR	68.75	18.75	15.63	12.50	6.25	7.29	5.21	3.13	4.17	3.13	3.13

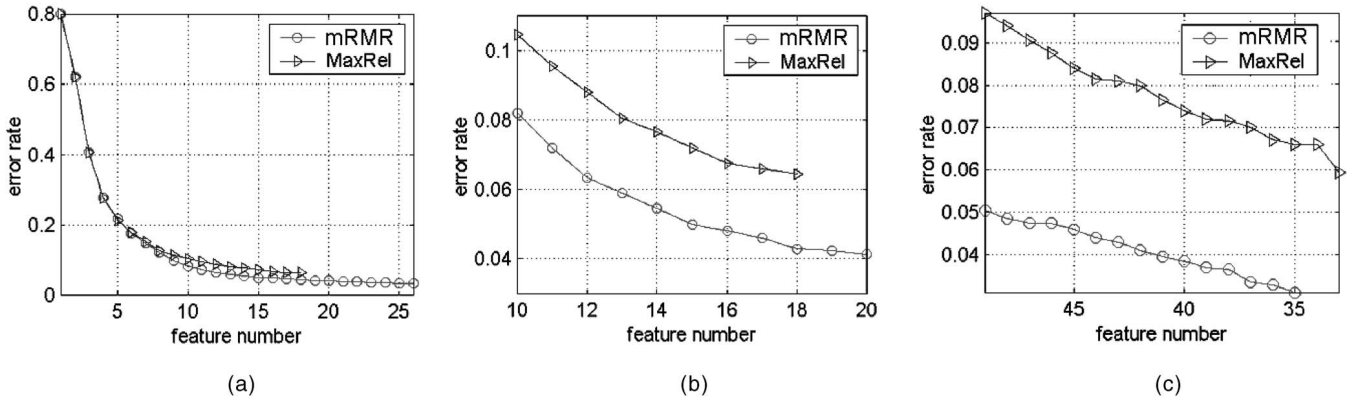


Fig. 5. The wrapper selection/classification results (HDR + NB). (a) Forward selection. (b) Forward selection (zoom-in). (c) Backward selection.

5.4 Comparison of Compact Feature Subsets Selected by mRMR and MaxRel

The results of incrementally selected candidate features have indicated that, with the same number of *sequential* features, mRMR feature set has more characteristic strength than MaxRel feature set. Here, we investigate that, given the same number of candidate features, whether the mRMR feature space is RM-characteristic and contains a more characterizing *nonsequential* feature subspace than the MaxRel feature space. This can be examined using the wrapper methods in Section 3.

Since the first 50 features lead to reasonably stable and small error for every data set and classification method we tested (see Figs. 3 and 4 and Tables 2 and 3), we used the first 50 features selected by MaxRel and mRMR as the candidate features.

Both forward and backward selection wrappers were used to search for the optimal subset of features. If the candidate feature space of mRMR is RM-characteristic than that of MaxRel, wrappers should be able to find combinations of mRMR features that correspond to better classification accuracy.

As an example, Fig. 5 shows the classification error rates of optimal feature subsets selected by wrappers for HDR data set and NB classifier. Figs. 5a and 5b (the zoom-in

view of Fig. 5a) clearly show that forward-selection-wrapper can consistently find a significantly better subset of features from the mRMR candidate feature set than from the MaxRel candidate feature set. This indicates that the mRMR candidate feature set is RM-characteristic for the forward-selection of wrapper. For MaxRel, wrapper obtains the lowest error 6.45 percent by selecting 18 features; more features will increase the error (thus, the wrapper selection is terminated). In contrast, by selecting 18 mRMR features, wrapper has ~ 4 percent classification error; it achieves even lower classification error with more mRMR features, e.g., 3.2 percent error for 26 features.

Fig. 5c shows that the backward-selection-wrapper also finds superior subsets from the candidate features generated by mRMR. Such feature subset always lead to significantly lower error rate than the subset selected from MaxRel candidate features. This indicates the space of candidate features generated by mRMR does embed a subspace in which the data samples can be more easily classified.

Table 4 summarizes the results obtained for all four data sets and three classifiers. Obviously, similar to the HDR data, for almost all combinations of data sets, wrapper selection methods, and classifiers, lower error rates are attained from the mRMR candidate features, indicating that wrappers find

TABLE 4
Comparison of Different Wrapper Selection Results (Lowest Error Rate (%))

Data set	Wrapper	NB		SVM		LDA	
		MaxRel	mRMR	MaxRel	mRMR	MaxRel	mRMR
HDR	Forward	6.45	3.20	5.50	3.45	6.80	4.05
	Backward	5.95	3.10	4.55	2.85	6.90	4.00
ARR	Forward	18.81	17.86	20.95	19.29	19.76	18.10
	Backward	23.10	17.86	20.48	19.52	20.48	18.33
NCI	Forward	26.67	13.33	25.00	21.67	31.67	33.33
	Backward	20.00	15.00	18.33	13.33	30.00	25.00
LYM	Forward	6.25	5.21	6.25	2.08	6.25	2.08
	Backward	5.21	3.13	3.13	3.13	6.25	3.13

more characterizing feature subspaces from the mRMR features than from MaxRel features. We can conclude that mRMR candidate feature sets do cover a wider spectrum of the more characteristic features. (There are two exceptions in Table 4, for which the obtained feature subsets are comparable: 1) "NCI+LDA+Forward," where five mRMR features lead to 20 errors (33.33 percent) and seven MaxRel features lead to 19 errors (31.67 percent) and 2) "LYM+SVM+Backward," the same error (3.13 percent) is obtained.)

6 DISCUSSIONS

In our approach, we have stressed that a well-designed filter method, such as mRMR, can be used to enhance the wrapper feature selection, in achieving both high accuracy and fast speed. Our method uses an optimal first-order incremental selection to generate a candidate list of features that cover a wider spectrum of characteristic features. These candidate features have similar generalization strength on different classifiers (as seen in Figs. 3 and 4 and Tables 2 and 3). They facilitate effective computation of wrappers to find compact feature subsets with superior classification accuracy (as shown in Fig. 5 and Table 4). Our algorithm is especially useful for large-scale feature/variable selection problems where there are at least thousands of features/variables, such as medical morphometry [23], [8], gene selection [32], [31], [5], [12], etc.

Of note, the purpose of the mRMR approach studied in this paper is to maximize the dependency. This typically involves the computation of multivariate joint probability, which is nonetheless difficult and inaccurate. Combining both Max-Relevance and Min-Redundancy criteria, the mRMR incremental selection scheme provides a better way to maximize the dependency. In this case, the difficult problem of multivariate joint probability estimation is reduced to estimation of multiple bivariate probabilities (densities), which is much easier. Our comparison in Section 5.2 demonstrates that mRMR is a very good approximation scheme to Max-Dependency. In most situations, mRMR reduces the feature selection time dramatically for continuous features and improves the classification accuracy significantly. For data sets with a large number of samples, e.g., the HDR data set, the classification accuracy of mRMR is close to or better than that of Max-Dependency. We notice that the mRMR approach could also be applied to other domains where the similar heuristic algorithms are applicable to maximize the dependency of variables, such as

searching (learning) the locally optimal structures of Bayesian networks [24].

Our scheme of mRMR does not intend to select features that are independent of each other. Instead, at each step, it tries to select a feature that minimizes the redundancy and maximizes the relevance. For real data, the features selected in this way will have more or less correlation with each other. However, our analysis and experiments show that the joint effect of these features can lead to very good classification accuracy. A set of features that are completely independent of each other usually would be less optimal.

All of the feature selection methods used in this paper, including incremental search, forward, or backward selection, etc., are heuristic search methods. None of them can guarantee the global maximization of a criterion function. The fundamental problem is the difficulty in searching the whole space, as pointed out at the beginning of this paper. Additionally, questing the global optimum strictly might lead to data overfitting. On the contrary, mRMR seems to be a practical way to achieve superior classification accuracy in relatively low computational complexity.

Our experimental results show that, although, in general, more mRMR features will lead to a smaller classification error, the decrement of error might not be significant for each additional feature, or occasionally there could be fluctuation of classification errors. For example, in Fig. 3, the fifth mRMR feature seemingly has not led to a major reduction of the classification error produced with the first four features. Many factors count for these fluctuations. One cause is that additional features might be noisy. Another possible cause is that the mRMR scheme in (6) takes difference of the relevance term and the redundancy term. It is possible that one redundant feature also has relatively large relevance, so it could be selected as one of the top features. A greater penalty on the redundancy term would lessen this problem. A third possible cause is that the cross-validation method used might also introduce some fluctuations of the error curve. While a more detailed discussion on this fluctuation problem and other potential causes is beyond the scope of this paper, a way to solve this problem is to use other feature selectors to directly minimize the classification error and remove those potentially unneeded features, as what we do in the second stage of our algorithm. For example, by using wrappers in the second stage, the error curves thus obtained in Fig. 5 are much smoother than those obtained using the first-stage only in Fig. 3.

Our results show that for continuous data, density estimation works well for both mutual information

calculation and naive Bayes classifier. In earlier work, Kwak and Choi [17] used the density estimation approach to calculate the mutual information between an individual feature x_i and the target class c . We used a different approach based on direct Parzen-window approximation similar to [7]. Our results indicate that the estimated density and mutual information among continuous feature variables can be utilized to reduce the redundancy and improve the classification accuracy.

Finally, we notice that equation (6) is not the only possible mRMR scheme. Instead of combining relevance and redundancy terms using difference, we can consider quotient [5] or other more sophisticated schemes. The quotient-combination imposes a greater penalty on the redundancy. Empirically, it often leads to better classification accuracy than the difference-combination for candidate features. However, the joint effect of these features is less robust when some of them are eliminated; as a result, in the second-stage of feature selection using wrappers, the set of features induced from the quotient-combination would have a bigger size than that from the difference-combination. The mRMR paradigm can be better viewed as a general framework to effectively select features and allow all possibilities for more sophisticated or more powerful implementation schemes.

7 CONCLUSIONS

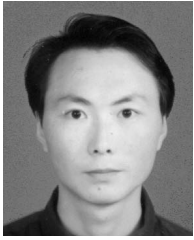
We present a theoretical analysis of the minimal-redundancy-maximal-relevance (mRMR) condition and show that it is equivalent to the maximal dependency condition for first-order feature selection. Our mRMR incremental selection scheme avoids the difficult multivariate density estimation in maximizing dependency. We also show that mRMR can be effectively combined with other feature selectors such as wrappers to find a very compact subset from candidate features at lower expense. Our comprehensive experiments on both discrete and continuous data sets and multiple types of classifiers demonstrate that the classification accuracy can be significantly improved based on mRMR feature selection.

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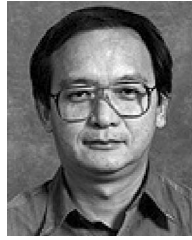
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Research

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Gene selection algorithm by combining reliefF and mRMR

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Abstract

Background: Gene expression data usually contains a large number of genes, but a small number of samples. Feature selection for gene expression data aims at finding a set of genes that best discriminate biological samples of different types. In this paper, we present a two-stage selection algorithm by combining ReliefF and mRMR: In the first stage, ReliefF is applied to find a candidate gene set; In the second stage, mRMR method is applied to directly and explicitly reduce redundancy for selecting a compact yet effective gene subset from the candidate set.

Results: We perform comprehensive experiments to compare the mRMR-ReliefF selection algorithm with ReliefF, mRMR and other feature selection methods using two classifiers as SVM and Naive Bayes, on seven different datasets. And we also provide all source codes and datasets for sharing with others.

Conclusion: The experimental results show that the mRMR-ReliefF gene selection algorithm is very effective.

Background

Gene expression refers to the level of production of protein molecules defined by a gene. Monitoring of gene expression is one of the most fundamental approach in genetics and molecular biology. The standard technique for measuring gene expression is to measure the mRNA instead of proteins, because mRNA sequences hybridize with their complementary RNA or DNA sequences while this property lacks in proteins. The DNA arrays, pioneered in [1,2], are novel technologies that are designed to measure gene expression of tens of thousands of genes in a single experiment. The ability of measuring gene expression

for a very large number of genes, covering the entire genome for some small organisms, raises the issue of characterizing cells in terms of gene expression, that is, using gene expression to determine the fate and functions of the cells. The most fundamental of the characterization problem is that of identifying a set of genes and its expression patterns that either characterize a certain cell state or predict a certain cell state in the future [3].

When the expression dataset contains multiple classes, the problem of classifying samples according to their gene expression becomes much more challenging, especially

when the number of classes exceeds five [4]. Moreover, the special characteristics of expression data adds more challenge to the classification problem. Expression data usually contains a large number of genes (in thousands) and a small number of experiments (in dozens). In machine learning terminology, these datasets are usually of very high dimensions with undersized samples. In microarray data analysis, many gene selection methods have been proposed to reduce the data dimensionality [5].

Gene selection aims to find a set of genes that best discriminate biological samples of different types. The selected genes are "biomarkers", and they form "marker panel" for analysis. In general, two types of gene selection methods have been studied in the literature: filter methods [6] and wrapper methods [7]. As pointed out in [8], the essential differences between the two methods are:

(1) that a wrapper method makes use of the algorithm that will be used to build the final classifier while a filter method does not, and

(2) that a wrapper method uses cross validation to compare the performance of the final classifier and searches for an optimal subset while a filter method uses simple statistics computed from the empirical distribution to select attribute subset.

Wrapper methods could perform better but would require much more computational costs than filter methods. Most gene selection schemes are based on binary discrimination using rank-based schemes [9], such as information gain, which reduces the entropy of the class variables given the selected attributes. In expression data, many gene groups interact closely and gene interactions are important biologically and may contribute to class distinctions. However, the majority of the rank-based schemes assume the conditional independence of the attributes given the target variable and are thus not effective for problems involving much feature interaction [10].

In this paper, we present a two-stage selection algorithm by combining ReliefF [10] and mRMR [11]. ReliefF, a general and successful attribute estimator, is able to effectively provide quality estimates of attributes in problems with dependencies between attributes. mRMR (minimal-redundancy-maximal-relevance) method selects genes that have the highest relevance with the target class and are also maximally dissimilar to each other. mRMR is computationally expensive. The integration of ReliefF and mRMR thus leads to an effective gene selection scheme. In the first stage, ReliefF is applied to find a candidate gene set. This filters out many unimportant genes and reduces the computational load for mRMR. In the second stage,

mRMR method is applied to directly and explicitly reduce redundancy and select a compact yet effective gene subset from the candidate set. We perform comprehensive experiments to compare the mRMR-ReliefF selection algorithm with ReliefF, mRMR and other feature selection methods using two classifiers on seven different datasets. The experimental results show that the mRMR-ReliefF gene selection is very effective.

Result and discussion

In this section, we perform comprehensive experiments to compare the mRMR-ReliefF selection algorithm with ReliefF, mRMR and other feature selection methods using two classifiers (Support Vector Machine (SVM) and Naive Bayes (NB)) on seven different datasets.

Datasets description

The datasets and their characteristics are summarized in Table 1.

- **ALL:** The ALL dataset [12] is a dataset that covers six subtypes of acute lymphoblastic leukemia: BCR (15), E2A (27), Hyperdip (64), MLL (20), T (43), and TEL (79). Here the numbers in the parentheses are the numbers of samples. The dataset is available at [13].

- **ARR:** The Arrhythmia (ARR) dataset contains 420 samples and 278 features with two classes [14].

- **GCM:** The GCM dataset [15] consists of 198 human tumor samples of fifteen types. breast (12), prostate (14), lung (12), colorectal (12), lymphoma (22), bladder (11), melanoma (10), uterus (10), leukemia (10), renal (11), pancreas (11), ovary (120), mesothelioma (11), CNS (20), and MET (9). The prediction accuracy of 78% is reported in [15] using one-versus-the rest SVM with all the genes.

- **HBC:** The HBC dataset consists of 22 hereditary breast cancer samples and was first studied in [16]. The dataset has three classes and can be downloaded at [17].

Table 1: The dataset description.

Dataset	# Samples	# Genes	# Classes
ALL	248	12558	6
ARR	420	278	2
GCM	198	16063	14
HBC	22	3226	3
LYM	62	4026	3
MLL	72	12582	3
NCI60	60	1123	9

- **LYM:** The Lymphoma dataset is a dataset of the three most prevalent adult lymphoid malignancies and available at [18] and it was first studied in [19].
- **MLL:** The MLL-leukemia dataset consists of three classes and can be downloaded at [20].
- **NCI60:** The NCI60 dataset was first studied in [21]. cDNA microarrays were used to examine the variation in gene expression among the 60 cell lines from the National Center Institute's anticancer drug screen. The dataset spans nine classes and can be downloaded at [17,22].

Note that in these datasets, the samples in each class is generally small, and unevenly distributed. This, together with the large number of classes, especially for NCI60, GCM, makes the classification task more complex.

Compare ReliefF, mRMR and mRMR-ReliefF algorithm

First we compare the mRMR-ReliefF algorithm with ReliefF and mRMR. We perform our comparisons using SVM and NB classifiers on the seven datasets. Both SVM and NB have been widely used in previous studies. Figure 1 and Figure 2 show the classification accuracy results as a function of the number of selected genes on the seven datasets respectively. In addition, because of mRMR is computationally expensive, using the program provided in [11], we could not obtain results for several datasets with a large number of genes, e.g., ALL and GCM. Thus in the figures, we only include the accuracy values for ReliefF and the mRMR-ReliefF algorithm and these values are all obtained via 10-fold cross validation.

Table 2 presents the detail of the accuracy values of applying SVM and NB classification on the top 30 selected genes, for some unavailable results which can not be computed by mRMR, we note them as "-". From the above comparative study, we observe that:

- The performance of mRMR algorithm is pulled down by its expensive computational cost, and it can not fulfill gene selection on the database with large features using the limited memory.
- Relief algorithm is not stable enough when only a small number of genes are selected. And when the number of selected genes is greater than 30, the variations of classification performance of both ReliefF and mRMR-ReliefF algorithms are generally small.
- The mRMR-ReliefF selection algorithm leads to significantly improved class predictions. With the same number of selected genes, the gene set obtained by the mRMR-ReliefF selection is more representative of the target class,

therefore leading to better class prediction or generalization property.

Comparison with other methods

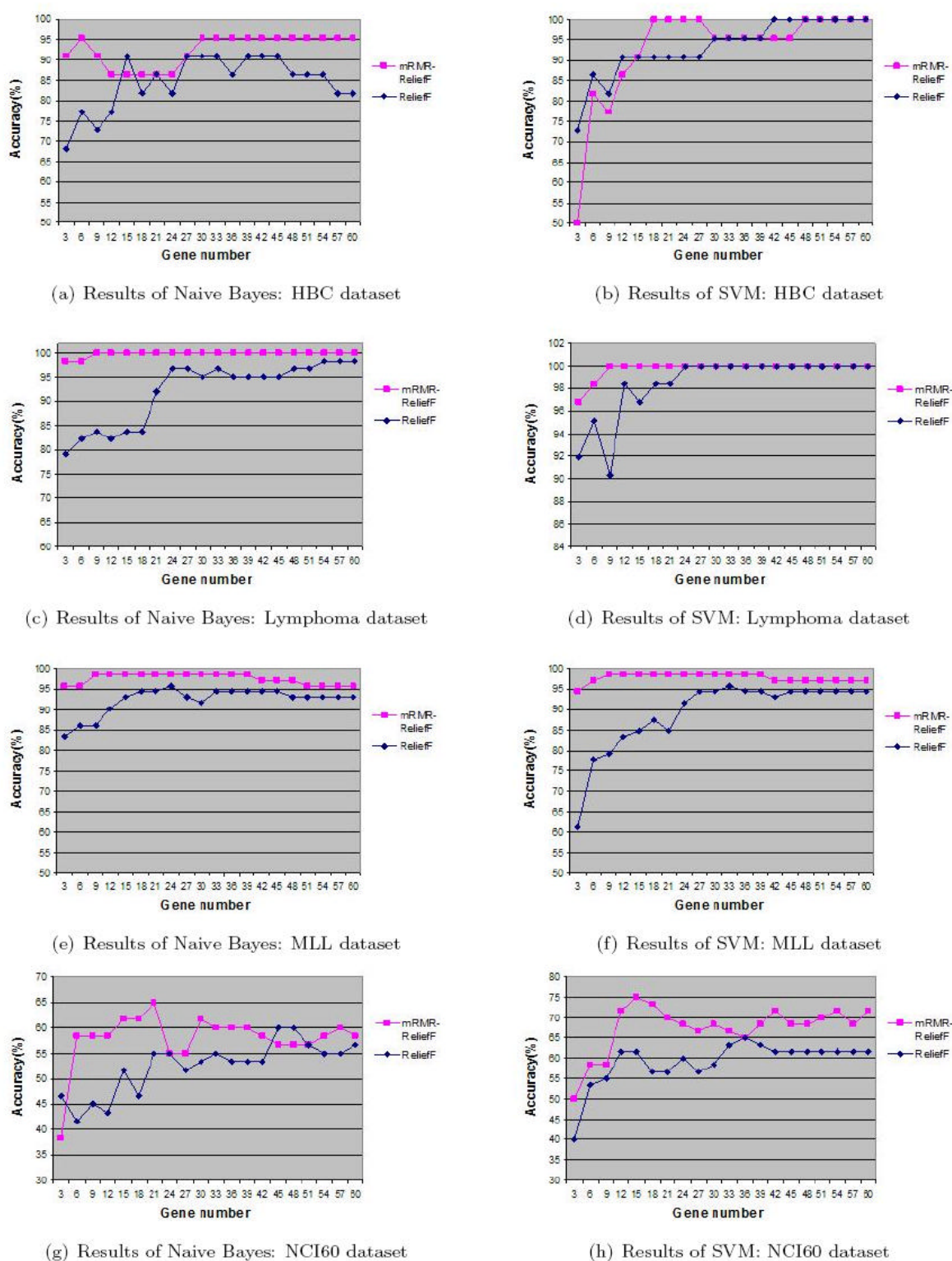
We also compare our mRMR-ReliefF selection algorithm with other gene selection algorithms, including Max-Relevance, Information Gain, Sum Minority, Twoing Rule, F-statistic [23], and GSNR [24]. Table 3 presents the classification accuracy comparison using SVM and NB classifier based on the selected genes using these six feature selection methods, when the number of selected gene is 30. From Table 3, we observe that:

- Gene selection improves class prediction. Note that the accuracy of SVM using feature selection generally outperforms that without feature selection. This implies that feature selection can effectively reduce the insignificant dimensions and noise to improve classification accuracy.
- The mRMR-ReliefF algorithm is shown to achieve better performance comparing with other gene selection algorithms on almost all datasets. The experimental comparisons demonstrate the effectiveness of the integration of ReliefF and mRMR.
- ReliefF achieves good performance on most of the data sets. Although its performance is not always as good as that of the mRMR-ReliefF algorithm. It outperforms mRMR, Maxrel, Sum Minority and partially wins information gain, twoing rule.
- Only a small number of genes are needed for classification purpose. In our experiments, the variations of the classification accuracy are small when the number of selected genes is greater than 30.

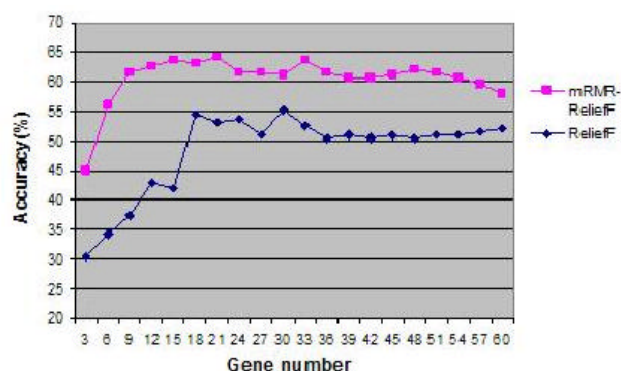
Software package

We have developed a software package for the above experiments, which includes: 1) The source codes for four feature selection algorithms including ReliefF, F-statistic, GNSR, and Relief-mRMR; 2) A MATLAB interface for Rankgene1.1 [5] which contains another eight feature selection measures; 3) A MATLAB interface for two well-known classification tools (e.g., LIBSVM and WEKA); 4) Programs for converting data formats; 5) The collection of all datasets used in the experiments. We hope it is a useful tool in gene expression analysis and feature selection.

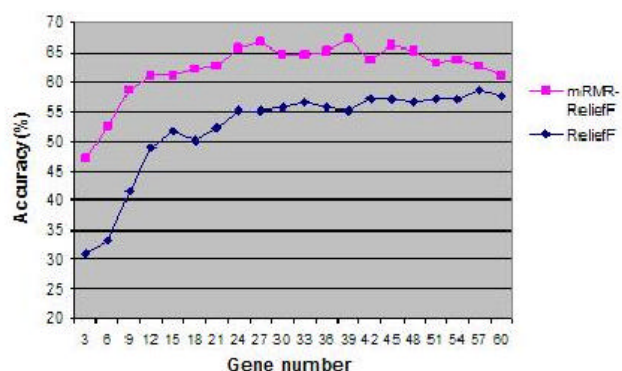
This package and all datasets can be downloaded from <http://www.cis.fiu.edu/~yzhan004/genesel.html>. All codes are implemented and tested in Matlab 7.0 and can be integrated into the Toolbox by adding its path to MATLAB search path.

**Figure 1**

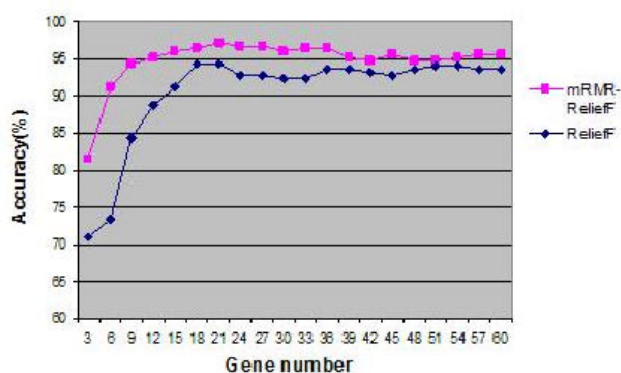
Comparison of ReliefF and mRMR-ReliefF algorithms I. This figure describes the two classifications (SVM and NB) results using 3 to 60 selected genes, for HBC, Lymphoma, MLL, and NCI60 datasets. From this figure, it is easy to know that in the same number of selected genes, the performance of mRMR-ReliefF algorithm is obviously better than ReliefF algorithm.



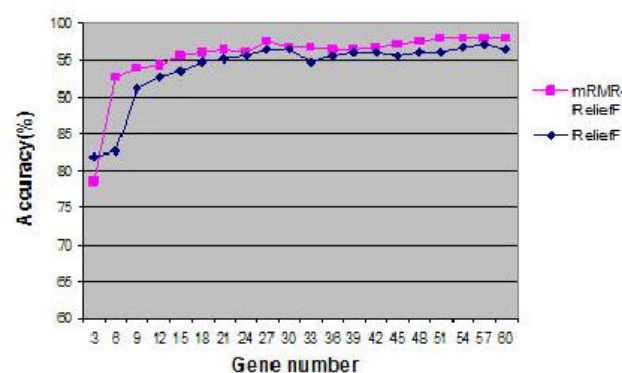
(a) Results of Naive Bayes: GCM dataset



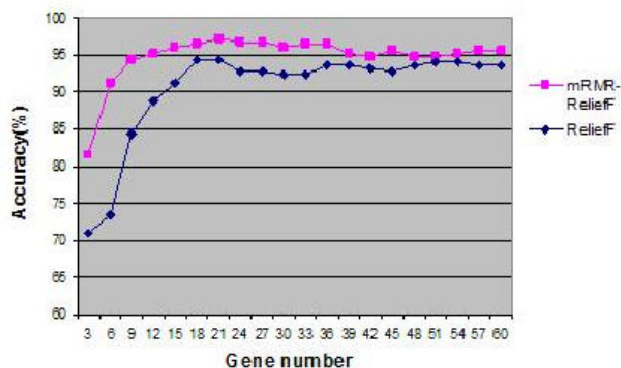
(b) Results of SVM: GCM dataset



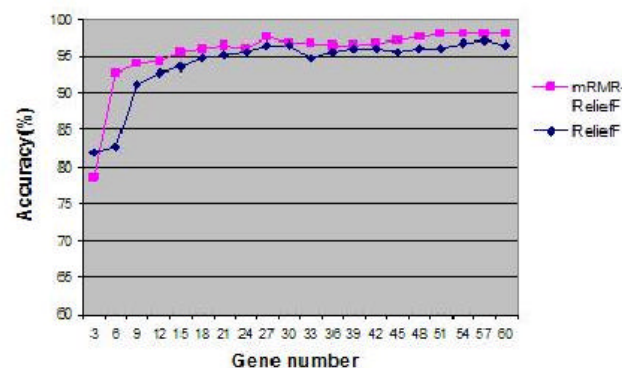
(c) Results of Naive Bayes: ALL dataset



(d) Results of SVM: ALL dataset



(e) Results of Naive Bayes: ALL dataset



(f) Results of SVM: ALL dataset

Figure 2

Comparison of ReliefF and mRMR-ReliefF algorithms II. This figure describes the two classifications (SVM and NB) results using 3 to 60 selected genes, for GCM, ALL, and ARR datasets. From this figure, it is easy to know that in the same number of selected genes, the performance of mRMR-ReliefF algorithm is obviously better than ReliefF algorithm.

Data structure and translation

This package supports consistent data formats. Each gene dataset is formatted as a MATLAB data structure file(.mat), in which a class label vector corresponds to a gene array. For any algorithm, the input is a .mat file, and the output

is an index vector for the selected genes. Furthermore, a utility is provided for converting the data from .csv file to .mat file. The command line is as follows.

```
csvtomat(Filename)
```

Table 2: The comparisons in ReliefF, mRMR and mRMR-ReliefF algorithms (gene number = 30)

Feature Selection Method	Classifier	ALL	ARR	LYM	HBC	NCI60	MLL	GCM
ReliefF	SVM	96.37%	79.29%	100%	95.45%	58.33%	94.44%	55.25%
	Naive Bayes	92.34%	75%	95.16%	90.91%	53.33%	91.67%	55.56%
mRMR	SVM	-	75.35%	100%	95.45%	53.33%	-	-
	Naive Bayes	-	73.21%	97.33%	87.51%	51.20%	-	-
mRMR-ReliefF	SVM	96.77%	81.43%	100%	95.45%	68.33%	98.61%	64.65%
	Naive Bayes	95.97%	79.05%	100%	95.45%	61.67%	98.61%	61.11%

where *Filename* is the name of .csv file. In the .csv file, the first column is the class label, the rest are gene variables. For .mat file, its structure can be shown as Figure 3:

We also provide the function to convert .mat file to .csv file as:

```
mattocsv(X, y, Filename)
```

where *X*, *y* are the matrix defined in .mat file and *Filename* is the .csv file as output file.

Implementation of gene selection algorithms

The command list to perform different gene selection algorithms is shown in Table 4, where *X* is a gene array, *y* is a class label vector, and *Topn* is the number of selected genes in current algorithm. For ReliefF function, *n* is the number of iterations, *K* is the number of neighbors to be selected, and *typed* is the data type; for the Rankgene function, and *T* is the method index which can be referenced in rankgene1.1.

Assistant tools for classification

To compare the performance of the gene selection algorithms, we also include two popular classification tools in this software package, which are the existing MATLAB version for LIBSVM [25] and a MATLAB Interface for WEKA [26]. For LIBSVM, there is already a ready-to-run plug-in for MATLAB. And we implement the function for calling WEKA. The command line for calling WEKA is shown as follows.

```
mattocsv(X, y, Filename)
```

```
Accuracy = wekaclassifier(Filename, Classifier)
```

where *Filename* is the name of the output .csv file, *X* is a gene array, *y* is a label vector, and *Classifier* is the parameter for classification method, such as Naive Bayes and J4.5 tree.

Conclusion

In this paper, we present an mRMR-ReliefF selection algorithm by combining ReliefF and mRMR. ReliefF is able to effectively provide quality estimates of attributes in problems with dependencies between attributes and mRMR

Table 3: The comparisons in seven gene selection methods (gene number = 30).

Feature Selection Method	Classifier	ALL	ARR	LYM	HBC	NCI60	MLL	GCM
No feature sel	SVM	91.94%	51.04%	95.16%	77.27%	63.33%	97.22%	51.52%
	Naive Bayes	85.23%	49.57%	95.04%	70.11%	45.22%	93.13%	40.33%
mRMR-ReliefF	SVM	96.77%	81.43%	100%	95.45%	68.33%	98.61%	64.65%
	Naive Bayes	95.97%	79.05%	100%	95.45%	61.67%	98.61%	61.11%
Maxrel	SVM	89.11%	74.53%	100%	72.73%	51.67%	77.78%	60.61%
	Naive Bayes	88.71%	73.49%	100%	63.64%	48.33%	80.56%	46.97%
Information Gain	SVM	97.58%	80.13%	98.39%	100%	61.67%	98.67%	46.67%
	Naive Bayes	92.74%	77.21%	93.55%	86.38%	60%	97.22%	47.47%
Sum Minority	SVM	93.95%	76.42%	98.39%	95.45%	55%	90.28%	55.05%
	Naive Bayes	91.13%	74.32%	95.16%	81.82%	46.67%	91.67%	49.49%
Twoing Rule	SVM	96.77%	79.37%	98.39%	90.91%	61.67%	97.22%	45.96%
	Naive Bayes	90.32%	72.19%	93.55%	86.36%	45%	95.83%	46.46%
F-statistic	SVM	97.17%	67.12%	96.77%	90.91%	63.33%	77.22%	39.10%
	Naive Bayes	80.27%	71.55%	98.52%	85.41%	60.15%	80.13%	39.81%
GSNR	SVM	93.18%	77.24%	100%	95.45%	63.37%	90.25%	40.74%
	Naive Bayes	90.11%	70.43%	100%	85.65%	58.25%	87.22%	39.81%

This table shows the classification results based on the 30 genes, which are selected from 7 different datasets using seven feature selection methods, named mRMR-ReliefF, Maxrel, information gain, sum minority, twoing rule, F-statistic, GSNR.

Input: Gene variables and labels

Output: W for the gene rank

Set all weights $W := 0$;

foreach Iteration n **do**

 Randomly select an instance Ins_m ;

 Find K nearest hits H ;

foreach class $c \neq Label_m$ **do**

 | from class c find K nearest misses M_c ;

end

foreach g_i **do**

 | Update W_i ;

end

end

Figure 3

Description of ReliefF algorithm.

method selects genes that have the highest relevance with the target class and are also maximally dissimilar to each other. The integration of ReliefF and mRMR thus leads to an effective gene selection scheme: In the first stage, ReliefF is applied to find a candidate gene set; In the second stage, mRMR is applied to select a compact yet effective gene subset from the candidate set.

Comprehensive experiments are conducted to compare the mRMR-ReliefF selection algorithm with ReliefF, mRMR and other feature selection methods using two classifiers on seven different datasets. The experimental results show that the mRMR-ReliefF gene selection is very effective. In addition, we also developed a software package to help other researches explore gene expression.

Methods

In this part, firstly, ReliefF and mRMR algorithms are discussed, then mRMR-ReliefF selection algorithm is presented, and finally, other six gene selection algorithms used to compare with our mRMR-ReliefF algorithm are introduced.

Table 4: MATLAB Command List For Gene Selection.

Algorithm Description	Command Line
ReliefF	$W = \text{reliefF}(X, y, n, K, \text{typed}, \text{Topn})$
F-statistic	$W = \text{Ftest}(X, y, \text{Topn})$
GNSR	$W = \text{Gsnr}(X, y, \text{Topn})$
ReliefF-mRMR	$W = \text{rm}(X, y, n, K, \text{Topn})$
Rankgene	$W = \text{rankgene}(X, y, T, \text{Topn})$

This table shows MATLAB commands for the feature selection algorithms, which are ReliefF, F-statistic, GNSR, ReliefF-mRMR, and all algorithms included in Rankgene.

ReliefF

ReliefF is a simple yet efficient procedure to estimate the quality of attributes in problems with strong dependencies between attributes [10]. In practice, ReliefF is usually applied in data pre-processing as a feature subset selection method.

The key idea of the ReliefF is to estimate the quality of genes according to how well their values distinguish between instances that are near to each other. Given a randomly selected instance Ins_m from class L , ReliefF searches for K of its nearest neighbors from the same class called nearest hits H , and also K nearest neighbors from each of the different classes, called nearest misses M . It then updates the quality estimation W_i for gene i based on their values for Ins_m, H, M . If instance Ins_m and those in H have different values on gene i , then the quality estimation W_i is decreased. On the other hand, if instance Ins_m and those in M have different values on the the gene i , then W_i is increased. The whole process is repeated n times which is set by users. The algorithm is shown in Figure 4 and updating W_i can use Equation 1:

$$W_i = W_i - \frac{\sum_{k=1}^K D_H}{n \cdot K} + \sum_{c=1}^{C-1} P_c \cdot \frac{\sum_{k=1}^K D_{M_c}}{n \cdot K} \quad (1)$$

where n_c is the number of instances in class c , D_H (or D_{M_c})

the sum of distance between the selected instance and each H (or M_c), P_c is the prior probability of class c .

Detailed discussions on ReliefF can be found in [10] and recently, it was shown that ReliefF is an on-line solution to a convex optimization problem, maximizing a margin-based algorithm [27].

$\mathbf{v} =$

```

X: [62x4026 double]
y: [62x1 double]
name: 'Finite data set'
dim: 4026
num_data: 62

```

Figure 4

The data structure description for software package. X is the gene array with 62 genes and 4026 expression variables. y is the label for each gene.

mRMR

The mRMR (minimum redundancy maximum relevance) method [11] selects genes that have the highest relevance with the target class and are also minimally redundant, i.e., selects genes that are maximally dissimilar to each other. Given g_i which represents the gene i , and the class label c , their mutual information is defined in terms of their frequencies of appearances $p(g_i)$, $p(c)$, and $p(g_i, c)$ as follows.

$$I(g_i, c) = \int p(g_i, c) \ln \frac{p(g_i, c)}{p(g_i)p(c)} dg_i dc \quad (2)$$

The Maximum-Relevance method selects the top m genes in the descent order of $I(g_i, c)$, i.e. the best m individual features correlated to the class labels.

$$\max_S \frac{1}{|S|} \sum_{g_i \in S} I(g_i, c) \quad (3)$$

Although we can choose the top individual genes using Maximum-Relevance algorithm, it has been recognized that "the m best features are not the best m features" since the correlations among those top features may also be high [28]. In order to remove the redundancy among features, a Minimum-Redundancy criteria is introduced

$$\min_S \frac{1}{|S|^2} \sum_{g_i, g_j \in S} I(g_i, g_j) \quad (4)$$

where mutual information between each pair of genes is taken into consideration. The minimum-redundancy maximum-relevance (mRMR) feature selection framework combines both optimization criteria of Eqs.(3, 4).

A sequential incremental algorithm to solve the simultaneous optimizations of optimization criteria of Eqs.(3, 4) is given as the following. Suppose set G represents the set of genes and we already have S_{m-1} , the feature set with $m-1$ genes. Then the task is to select the m -th feature from the set $\{G - S_{m-1}\}$. This feature is selected by maximizing the *single-variable relevance minus redundancy* function

$$\max_{g_j \in G - S_{m-1}} [I(g_j, c) - \frac{1}{m-1} \sum_{g_i \in S_{m-1}} I(g_j, g_i)] \quad (5)$$

The m -th feature can also be selected by maximizing the *single-variable relevance divided-by redundancy* function

$$\max_{g_j \in G - S_{m-1}} [I(g_j, c) / \frac{1}{m-1} \sum_{g_i \in S_{m-1}} I(g_j, g_i)] \quad (6)$$

mRMR-ReliefF algorithm

As we mentioned before, ReliefF is a general and successful attribute estimator and is able to effectively provide

quality estimates of attributes in problems with dependencies between attributes. However, ReliefF does not explicitly reduce the redundancy in selected genes. mRMR selects genes that have the highest relevance with the target class and are also maximally dissimilar to each other. However, mRMR is computationally expensive. For example, using the mRMR program provided in [11], we could not obtain results for several datasets with a large number of genes, e.g., ALL and GCM. The integration of ReliefF and mRMR thus leads to an effective gene selection scheme.

We can view the *quality estimation* W_i in ReliefF as maximizing the relevance score. Thus we can view the standard ReliefF algorithm as maximizing the relevance score:

$$\max_S \frac{1}{|S|} \sum_{g_i \in S} W_i \quad (7)$$

Thus our mRMR-ReliefF algorithm selection criteria becomes

$$\max_{g_j \in G - S_{m-1}} W_j - \frac{1}{m-1} \sum_{g_i \in S_{m-1}} |C(g_j, g_i)| \quad (8)$$

or

$$\max_{g_j \in G - S_{m-1}} W_j / \frac{1}{m-1} \sum_{g_i \in S_{m-1}} |C(g_j, g_i)| \quad (9)$$

where $C(g_j, g_i)$ is the Pearson correlation coefficient.

Our mRMR-ReliefF algorithm works as follows: In the first stage, ReliefF is applied to find a candidate gene set. This filters out many unimportant genes and reduces the computational load for mRMR. In the second stage, mRMR method is applied to directly and explicitly reduce redundancy and select a compact yet effective gene subset from the candidate set.

In our experiments, ReliefF is first used to choose 150 genes as the candidate set. from the all gene data. mRMR algorithm is then applied to select the final subset.

Other gene selection algorithms

In this part, we introduce six other gene selection algorithms which are mentioned in the chapter of "Result and discussion", which are named Max-Relevance, Information Gain, Sum Minority, Twoing Rule, F-statistic [23], and GSNR [24]. These methods have been reported in previous work. The first four methods have been used either in machine learning (information gain) or in statistical learning theory (twoing rule and sum minority), and all of them measure the effectiveness of a feature by evaluating the strength of class prediction when the prediction is

made by splitting it into two regions, the high region and the low region, by considering all possible split points [5]. More detailed descriptions on these methods can be found in [5].

F-statistic is chosen to score the relevance between the genes and the classification variable. The F-statistic of gene i in C classes has the following form [23]:

$$W_i = \frac{\sum_{c=1}^C n_c \cdot (\bar{g}_{ic} - \bar{g}_i) / (C-1)}{\sum_{c=1}^C \{(n_c-1) \sum_{j=1}^{n_c} (g_{jic} - \bar{g}_{ic})^2 / n_c\} / (n-C)}$$

where C is the number of classes, $\bar{g}_i$ is the mean of gene i variables, n_c is the number of samples in class c , $\bar{g}_{ic}$ is the mean of gene i in class c , and g_{jic} is sample j in gene i value in class c .

As to GSNR, it has been proposed and used in [24]. GSNR is a measure of the ratio between inter-group and intra-group variations. Higher GSNR values indicate higher discrimination power for the gene. The GSNR value for gene i is given by:

$$W_i = \frac{\sum_{c=1}^C \bar{g}_{ic} - \sum_{c=1}^C \bar{g}_i / C}{\sum_{c=1}^C \sum_{j=1}^{n_c} |g_{jic} - \bar{g}_{ic}| / n_c} \quad (11)$$

Both F-statistic and GSNR select m genes in the descent order of W_i , and the best subset of genes is satisfied the following description:

$$\max_s \frac{1}{|S|} \sum_{g_i \in S} W_i \quad (12)$$

Competing interests

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Authors' contributions

T. Li and C. Ding initialized the idea and supervised the project. Y. Zhang implemented the algorithms, developed the software, performed experimental comparisons, and built the website. All authors have read and approved the manuscript.

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